

ORIGINAL RESEARCH

Global, Regional, and National Burden of Myocarditis and Its Impairment, 1990-2023

GBD 2023 Myocarditis Collaborators

ABSTRACT

BACKGROUND The impact of recent trends in the burden of myocarditis has not been reported in detail.

OBJECTIVES This study aims to estimate the global, regional, and national burden of myocarditis from 1990 to 2023, using data from the GBD (Global Burden of Disease, Injuries, and Risk Factors Study) 2023.

METHODS As part of the GBD 2023, the authors used all available administrative health facility and vital registration data to estimate myocarditis prevalence, incidence, disability-adjusted life years (DALYs), mortality, and mortality-to-incidence ratio (MIR) for every country, stratified by age, sex, and quintiles of a Socio-demographic Index (SDI). The MIR represents the relative mortality burden compared with disease incidence at the population level. The prevalence of heart failure attributable to myocarditis was estimated using a model-based attribution approach.

RESULTS In 2023, the global age-standardized rates of prevalence, incidence, DALY, and mortality of myocarditis were 4.84 (95% uncertainty interval [UI]: 3.98-5.80), 12.46 (95% UI: 9.86-15.70), 8.41 (95% UI: 5.63-12.59) and 0.21 (95% UI: 0.14-0.29) per 100,000 population, respectively. Although global age-standardized prevalence and incidence rates remained stable from 1990 to 2023, the absolute number of incident cases increased by 54.28% from 0.67 million (95% UI: 0.52-0.85 million) to 1.04 million (95% UI: 0.83-1.32 million). Age-standardized DALYs and mortality rates decreased by 62.48% and 60.00%, respectively, over the same period. The MIR decreased across all age groups from 1990 to 2023 and remained highest among the youngest and oldest populations, with the ratio in those <5 years of age declining from 17.82% in 1990 to 6.32% in 2023. Across SDI strata, incidence rates remained stable from 1990 to 2023, while mortality rates declined in all quintiles and remained lowest in high SDI regions, whereas the MIR decreased from 2.33 in 1990 to 1.30 in 2023. In 2023, prevalence was highest in the high SDI quintile (6.30 [95% UI: 5.15-7.48] per 100,000 population) and lowest in the low-middle SDI quintile (4.10 [95% UI: 3.38-5.02]), reflecting disparities aligned with health resource availability. Age-standardized prevalence rates of heart failure attributable to myocarditis declined from 2.33 (95% UI: 1.90-2.90) per 100,000 population in 1990 to 1.76 (95% UI: 1.41-2.29) in 2023, with a U-shaped distribution across age groups showing the highest burden in those <5 years of age and >75 years of age in 2023.

CONCLUSIONS Although age-adjusted DALYs and mortality rates of myocarditis decreased globally from 1990 to 2023, incidence and prevalence remained stable, suggesting potential improvements in outcomes over time. However, absolute burden has grown due to population growth and aging, and important disparities persist across age groups and SDI strata. These findings highlight the need for targeted strategies to improve early diagnosis and reduce the burden of myocarditis, particularly in vulnerable populations and lower-resource settings, while being interpreted in the context of evolving diagnostic and coding practices. (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

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This manuscript is official IHME collaborator-lead paper “GBD 2023 Myocarditis Collaborators”.

ABBREVIATIONS AND ACRONYMS

CMR = cardiac magnetic resonance

DALY = disability-adjusted life year

EMB = endomyocardial biopsy

MIR = mortality-to-incidence ratio

SDI = Socio-demographic Index

UI = uncertainty interval

YLD = year lived with disability

YLL = year of life lost

Myocarditis often has a favorable outcome without long term disability; however sudden death and fulminant heart failure disproportionately affect children and older persons.^{1,2} Previous studies reported that acute myocarditis accounted for 3% to 12% of cases of sudden cardiac death in young adults.³ Furthermore, nearly half of the cases of dilated cardiomyopathy in children, where the cause is identified, was attributed to myocarditis.⁴ As the use of immune checkpoint inhibitors has increased, immune-related adverse events have

increasingly contributed to myocarditis rates in older adults. Left untreated, myocarditis can lead to irreversible myocardial injury.

The causes of myocarditis also vary by regions with different infective pathogens, autoimmune diseases, allergens, drug use, and genetic predispositions that can impact the regional risk of myocardial inflammation.^{1,5} Over the past decade, new factors including immune checkpoint inhibitors and genetic risk that contribute to myocarditis burden have been identified. Particularly during the COVID-19 pandemic, myocarditis was reported as both an early and long-term cardiovascular complication.⁶ Furthermore, myocarditis following COVID-19 mRNA vaccination evoked concern about vaccine safety; although, the impact on population risk remains an area of controversy.⁷

Previous studies have attempted to estimate the incidence trend of myocarditis, but the majority of these studies have focused on high-income countries with greater access to diagnostic tests, including cardiac magnetic resonance (CMR) and endomyocardial biopsy (EMB).^{8,9} Estimating the population-level burden of myocarditis poses analytical challenges due to diagnostic resources. Diagnostic capacity, clinical thresholds for case identification, and coding practices can vary between countries and over time. ICD (International Classification of Disease)-coded health facility records and death certificates remain an important source of information for population-level surveillance of myocarditis but have not been systematically evaluated.

As part of the GBD (Global Burden of Disease, Injuries, and Risk Factors Study) 2023, this analysis estimates the global and national burden of myocarditis with a focus on mortality-to-incidence ratios (MIRs) to clarify disease severity and the demographic drivers of disease burden, including population growth and aging.

METHODS

OVERVIEW. The GBD 2023 provides a comprehensive assessment of disease burden by quantifying health loss from 375 diseases across 204 countries and territories from 1990 to 2023. It generates standardized GBD metrics including incidence, prevalence, cause-specific mortality, years of life lost (YLLs), years lived with disability (YLDs), and disability-adjusted life years (DALYs).^{10,11} Age-standardized estimates were generated using the GBD world population age standard. Data sources can be accessed through the GHDx, a publicly searchable database of model inputs. Estimates of incidence and prevalence were derived from DisMod-MR 2.1 models using 1,000 posterior draws, consistent with standard GBD methodology. Uncertainty intervals (UIs) for these estimates were defined by the 2.5th and 97.5th of the 1,000 draws. For subsequent calculations of YLDs and DALYs, draws were down sampled to 250 using a correlated percentile-based approach that preserves the shape and width of the posterior distribution while reducing computational burden. Analyses were conducted utilizing Python version 3.10.4 and R version 4.2.1.^{10,11} Detailed methods are outlined in the [Supplemental Methods](#); a summary of methods follows. GBD 2023 adheres to the GATHER (Guidelines for Accurate and Transparent Health Estimates Reporting) checklist.¹² The completed GATHER checklist is available in [Supplemental Table 1](#). This study used de-identified data and was approved by the University of Washington Institutional Review Board (approval number: 9060).

CASE DEFINITION. Myocarditis refers to pathological inflammation of the myocardium, which can arise from various factors, including viral infections, autoimmune conditions, and other nonischemic causes, potentially leading to a reduced capacity of the heart to effectively pump blood throughout the

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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body. In this study, acute myocarditis was defined according to clinical diagnosis, based on ICD codes in administrative health facility data. ICD codes in vital registration (mortality) records were mapped to the GBD cause list (B33.2, I40-I41.8, I51.4-I51.6 for ICD-10 and 422-422.9, 429, and 429.1 for ICD-9).¹³

Heart failure is the major clinical consequence contributing to DALY in this GBD analysis. In this study, prevalent heart failure due to myocarditis was estimated separately from incident acute myocarditis in order to estimate its long-term effects. Heart failure was defined as a clinician-diagnosed case based on structured criteria such as the Framingham or European Society of Cardiology criteria. Cases included stage C or D according to the Universal Definition of Heart Failure. Inputs for the estimation of heart failure included cases documented in administrative health facility data and published population-based cohort studies.¹⁴

MORTALITY ESTIMATION. Mortality from myocarditis was modeled using a CODEm (cause of death ensemble model) approach using ICD-coded vital registration records as input data. Records coded to intermediate, implausible, or unspecified causes of death, including heart failure, were redistributed to valid underlying causes of death via reclassification algorithms designed to address comparability of mortality data sources across locations. CODEm produced an ensemble of a set of statistical timeseries submodels using 3 location-specific predictive covariates chosen a priori: 1) the health care access and quality index; 2) log-transformed lag distributed income per capita; and 3) a Socio-demographic Index (SDI). Submodels were combined into an ensemble using out-of-sample predictive validity weighting to produce a final estimate of cause-specific mortality.^{10,11}

DECOMPOSITION ANALYSIS. To investigate changes in myocarditis DALYs attributable to demographic and epidemiological factors, we decomposed the total change in DALYs between 1990 and 2023 into 3 components: 1) population growth; 2) population aging; and 3) epidemiological change, using a Das Gupta decomposition approach.¹⁵ DALYs were calculated as:

$$DALY_y = \sum_{i=1}^I (P_y \times A_{i,y} \times R_{i,y})$$

Where total DALYs in year (y) were modeled as a function of population size, age structure, and age-specific DALY rates. Specifically, P_y denotes the total population in year, $A_{i,y}$ represents the proportion

of the population in age group i , $R_{i,y}$ indicates the age-specific DALY rate, and I denotes the total number of age groups.

The change in DALYs between 1990 and 2023 was decomposed as:

$$\Delta DALY = DALY_{2023} - DALY_{1990} = \Delta P + \Delta A + \Delta R$$

where ΔP represents the contribution of population growth, ΔA reflects changes in age structure, and ΔR captures epidemiological changes in age-specific DALY rates. To avoid order-dependent bias, contributions of each component were estimated using a symmetric decomposition framework based on counterfactual scenarios.

MORBIDITY ESTIMATION. Estimation of incidence and prevalence of acute myocarditis and heart failure due to myocarditis was performed using DisMod-MR 2.1 (disease model–Bayesian meta-regression) state-transition disease modeling software.^{10,11,16} Coherence among disease parameters was ensured through differential equations with appropriate boundary conditions, borrowing strength from a geographic hierarchy of priors handed down to increasing levels of granularity to inform estimates in data-sparse locations. Acute myocarditis was estimated using a single DisMod-MR 2.1 model. Input data included hospital admission data identifying cases of acute myocarditis. These sources were adjusted for potential biases, including readmissions, transitions from primary to diagnosis recorded in any position of the record, and differences between inpatient and outpatient utilization. Correction factors derived from U.S. claims data were applied for these adjustments.^{10,11} Within the Bayesian meta-regression framework of DisMod-MR, the remission rate was assigned a value prior ranging from 3 to 5 per year, corresponding to an assumed average duration of ~3 months of acute illness. The lay description for acute myocarditis—“has a fever and aches, and feels weak, which causes some difficulty with daily activities”—reflects a mild, transient health state. Given the relative rarity of fulminant myocarditis compared with less severe presentations, and in line with the study objective of estimating nonfatal prevalence, excess mortality was specified as 0 across all age groups.

Heart failure due to myocarditis, modeled to be the long-term effects of a more severe case of myocarditis, was estimated using a multistage process. First, the total prevalence of heart failure from all causes was estimated using data obtained from a systematic review of the published reports along

with health facility data, adjusted as described earlier.¹⁷ In addition, administrative health facility data were corrected for systematic bias compared to population-based studies using MR-BRT (meta-regression–Bayesian, regularized, trimmed) to estimate adjustment factors.¹⁸ Second, heart failure prevalence was distributed among selected etiologies, including myocarditis. These etiologies were identified through a systematic approach incorporating 3 key methodological elements: 1) review of published reports to identify diseases with established biological mechanisms leading to heart failure; 2) quantitative analysis of disease co-occurrence patterns in death certificates and inpatient records across 93 location-years; and 3) expert consultation. This process led to the identification of 27 underlying causes, including an “other” category to ensure comprehensive coverage. Third, we used the epidemiological relationship between prevalence, cause-specific mortality, and excess mortality to estimate the proportion of heart failure attributable to each cause. These proportions were then multiplied by the estimates of overall heart failure to give estimates of prevalent heart failure due to each cause. These estimates represent the population-level contribution of myocarditis to heart failure burden based on available epidemiological data and should not be interpreted as direct causal attribution at the individual level. Prevalence estimates were then divided into disease-specific severity levels, or sequela, based on analyses of the distribution of disease severity in a population-representative survey of the U.S. population.

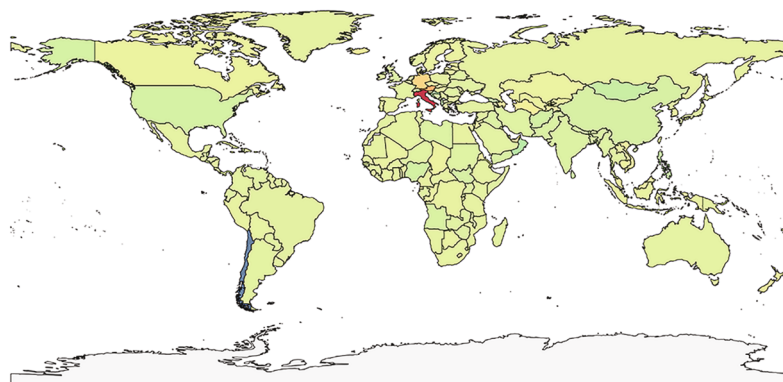
SUMMARY BURDEN MEASURES. Mortality estimates derived from CODEm were integrated with reference life tables to calculate estimates for YLLs. YLDs were estimated by pairing myocarditis-specific sequela (heart failure stage C and D) to the appropriate heart-failure and acute myocarditis health states and their disability weights. Disability weights used in GBD were constructed from surveys of the general population to reflect the impact of the health state on the population, based on review of health state lay descriptions. Prevalence for each sequela was multiplied by the corresponding disability weight to produce age-sex-location-year specific YLDs. Total YLDs were calculated by summing cause-specific YLDs. DALYs were calculated as the sum of YLLs and YLDs, with 1 DALY equating to 1 year of healthy life lost. Additionally, MIR was calculated as the ratio of age-specific mortality rate to age-specific incidence rate to provide a population-level indicator of relative mortality burden across settings.

RESULTS

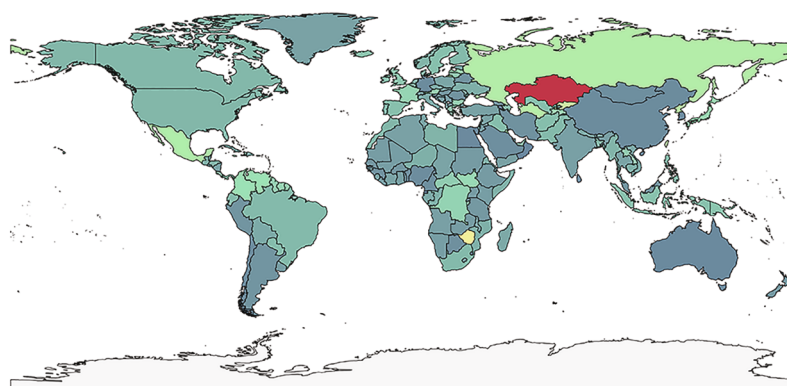
GLOBAL BURDEN OF MYOCARDITIS. Country levels of age-standardized prevalence, incidence, DALY, and mortality rates are depicted in [Figure 1](#) and [Supplemental Figure 1](#). In 2023, the global prevalence, incidence, DALY, and mortality rates of myocarditis were 4.84 (95% UI: 3.98-5.80), 12.46 (95% UI: 9.86-15.70), 8.41 (95% UI: 5.63-12.59), and 0.21 (95% UI: 0.14-0.29) per 100,000 population, respectively ([Table 1](#), [Supplemental Table 2](#)).

Globally, age-standardized myocarditis prevalence rates remained stable between 1990 and 2023 (5.55 [95% UI: 4.61-6.59] and 4.84 [95% UI: 3.98-5.80] per 100,000 population, respectively). During this period, the number of newly diagnosed myocarditis cases increased by 54.28% from 0.67 million (95% UI: 0.52-0.85 million) to 1.04 million (95% UI: 0.83-1.32 million) ([Figure 2](#), [Supplemental Figure 2](#)). The global age-standardized incidence rates also showed stability, with 13.11 (95% UI: 10.29-16.57) per 100,000 population in 1990 and 12.46 (95% UI: 9.86-15.70) in 2023. While the absolute number of deaths remained low and appeared stable (0.02 million [95% UI: 0.02-0.03 million] in 1990 and 0.02 million [95% UI: 0.01-0.02 million] in 2023), a relative reduction of 26.16% was observed. Conversely, global age-standardized mortality rates decreased by 60.00% from 0.51 (95% UI: 0.34-0.78) in 1990 to 0.21 (95% UI: 0.14-0.29) in 2023, respectively ([Supplemental Table 3](#)). The global trend of MIR by age groups, indicating the proportion of individuals who die from myocarditis among all newly diagnosed individuals, is presented in [Figure 2](#). Every age group showed a reduced MIR compared to 1990. Age groups exhibited U-shaped curves for the MIR, with the lowest ratio observed in the age group between 15 and 19 years. The group <5 years of age had a high MIR of 17.82% in 1990, decreasing to 6.32% in 2023. This elevated MIR in early childhood corresponded to a genuinely higher absolute mortality risk, as the mortality rate among children <5 years of age (0.278 [95% UI: 0.14-0.46] per 100,000 population) was higher than in the 5-9 years of age (0.046 [95% UI: 0.028-0.072] per 100,000 population) and 10 to 14 years of age (0.051 [95% UI: 0.032-0.078] per 100,000 population) groups, indicating that the elevated ratio in this group reflects a true increase in mortality risk and not simply a low incidence denominator.

BURDEN OF MYOCARDITIS ACCORDING TO AGE AND SEX. Subgroup analysis is presented in [Figure 3](#). Males had higher point estimates for myocarditis prevalence, incidence, DALYs, and mortality.

FIGURE 1 Overall Age-Standardized Incidence and Death Rate per 100,000 Population of Myocarditis, With Percent Change From 1990 to 2023, and Estimates for 2023**Percent Change of Incidence Rate, Both Sexes, Age-standardized, 1990 to 2023**

Percent Change 1990 to 2023	
1 Italy	19.95%
2 Austria	9.28%
3 Germany	5.34%
4 Maldives	4.40%
5 Malta	2.85%
...	
200 Albania	-2.55%
201 Croatia	-2.93%
202 Oman	-3.32%
203 Chile	-3.57%
204 Philippines	-14.15%

Percent Change of Death Rate, Both Sexes, Age-Standardized, 1990 to 2023

Percent Change 1990 to 2023	
1 Kazakhstan	204.8%
2 Zimbabwe	50.75%
3 Saint Vincent and the Grenadines	46.63%
4 Mauritius	34.42%
5 Kyrgyzstan	15.68%
...	
200 Guam	-78.64%
201 Oman	-81.38%
202 Equatorial Guinea	-82.52%
203 Monaco	-84.29%
204 Armenia	-89.7%

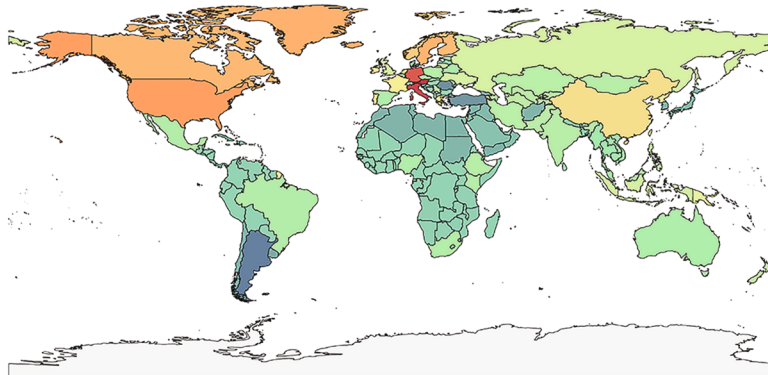
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However, age-standardized estimates were similar between the sexes. The prevalence rates per 100,000 population in males were similar (5.65 [95% UI: 4.64-6.81]) to those in females (4.01 [95% UI: 3.29-4.77]), as were the incidence rates (males: 14.89 [95% UI: 11.70-18.86]; females: 10.03 [95% UI: 7.98-12.57]), DALY rates (males: 10.13 [95% UI: 6.11-17.23]; females: 6.68 [95% UI: 3.38-13.71]), and mortality rates (males: 0.24 [95% UI: 0.15-0.37]; females: 0.17 [95% UI: 0.09-0.31]) (Supplemental Table 4).

BURDEN OF MYOCARDITIS ACCORDING TO SDI.

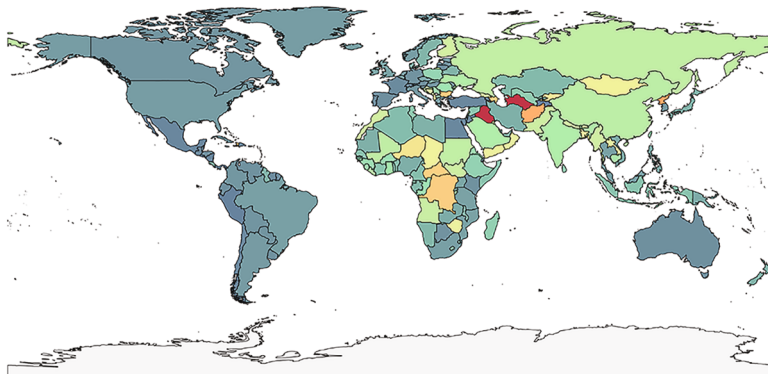
Age-standardized incidence rates remained relatively

stable over time and were broadly similar across all SDI quintiles, with modest fluctuations observed around periods aligning with the implementation of troponin testing and expansion of CMR (Figure 4). In contrast, age-standardized mortality rates showed a consistent decline from 1990 to 2023 across all SDI groups, with the lowest rates observed in high SDI countries throughout the study period. Despite overall reductions, disparities persisted, with low and low-middle SDI countries maintaining higher mortality rates compared to higher SDI regions. Consistent with these patterns, high SDI countries

FIGURE 1 Continued**Incidence Rate (Per 100,000 Population), Both Sexes, Age-Standardized, 2023**

2023 Incidence Rate (100,000 Population)	
1 Austria	21.17
2 Italy	20.62
3 Germany	19.85
4 United States of America	17.42
5 Greenland	16.93
...	
200 Uruguay	9.64
201 Georgia	9.57
202 Turkey	8.99
203 Romania	8.80
204 Argentina	8.05

9 12 15 18 21

Death Rate (Per 100,000 Population), Both Sexes, Age-Standardized, 2023

2023 Death Rate (100,000 Population)	
1 Turkmenistan	0.93
2 Iraq	0.93
3 Democratic People's Republic of Korea	0.65
4 Afghanistan	0.63
5 Central African Republic	0.58
...	
200 Puerto Rico	0.013
201 Saint Kitts and Nevis	0.011
202 Tajikistan	0.006
203 Micronesia (Federated States of)	0.004
204 Cook Islands	0.001

0.25 0.50 0.75

exhibited the lowest DALY and mortality rates (DALY rate: 5.31 [95% UI: 4.38-6.20] per 100,000 population; mortality rate: 0.13 [95% UI: 0.11-0.16]), while prevalence was higher in high SDI countries (low SDI: 4.25 [95% UI: 3.50-5.16] per 100,000 population; low-middle SDI: 4.10 [95% UI: 3.38-5.02] per 100,000 population; middle SDI: 4.17 [95% UI: 3.38-5.04] per 100,000 population; high-middle SDI: 5.03 [95% UI: 4.11-6.01] per 100,000 population; high SDI: 6.30 [95% UI: 5.15-7.48] per 100,000 population). The MIR declined steadily across all SDI quintiles from 1990 to 2023. In high SDI regions, the ratio decreased from 2.33 in 1990 to 1.30 in 2023. From 1990 to 2023, MIR remained consistently highest in low SDI regions and

lowest in high SDI regions, although the gap narrowed over time.

From 1990 to 2023, distinct epidemiological patterns were observed across SDI levels (Figure 4). After 2020, high, high-middle, and middle SDI countries experienced increasing age-standardized prevalence rates (high SDI: 6.25 [95% UI: 5.23-7.23] per 100,000 population in 2020 and 6.30 [95% UI: 5.15-7.48] in 2023; high-middle SDI: 4.97 [95% UI: 4.11-5.91] in 2020 and 5.03 [95% UI: 4.11-6.01] in 2023; middle SDI: 4.14 [95% UI: 3.41-4.99] in 2020 and 4.17 [95% UI: 3.38-5.04] in 2023) and age-standardized incidence rates (high SDI: 13.22 [95% UI: 10.73-16.02] per 100,000 population in 2020 and 13.75 [95% UI:

TABLE 1 Absolute Numbers (Thousands) and Age-Standardized Rates (per 100,000 People) of Prevalence, Incidence, DALYs, and Deaths of Myocarditis in 2023, and Percentage Change for 1990-2023

	Prevalence (95% UI)		Incidence (95% UI)		DALYs (95% UI)		Deaths (95% UI)	
	2023	Percent change, 1990-2023	2023	Percent change, 1990-2023	2023	Percent change, 1990-2023	2023	Percent change, 1990-2023
Global (overall)								
Absolute number	389.584 (321.411-466.573)	32.105 (26.517-39.533)	1,036.19 (826.447-1,315.627)	54.282 (44.663-63.577)	649.197 (439.752-968.650)	-46.09 (-66.496 to -16.946)	16.913 (11.282-24.091)	-26.157 (-49.55 to 13.298)
Age-standardized rate	4.837 (3.976-5.802)	-12.788 (-14.689 to -10.812)	12.463 (9.861-15.701)	-4.9 (-5.85 to -3.862)	8.414 (5.632-12.585)	-62.483 (-76.66 to -43)	0.205 (0.138-0.293)	-60.004 (-72.705 to -38.849)
Sex								
Male								
Absolute number	226.651 (185.479-273.019)	30.694 (25.398-37.412)	614.495 (484.432-785.635)	50.13 (41.188-58.968)	386.944 (238.838-650.766)	-48.901 (-72.468 to -9.503)	9.205 (6.029-14.158)	-33.712 (-61.836 to 19.95)
Age-standardized rate	5.653 (4.644-6.809)	-13.006 (-14.805 to -11.161)	14.888 (11.699-18.863)	-6.557 (-7.771 to -5.329)	10.131 (6.109-17.23)	-64.119 (-80.713 to -36.547)	0.241 (0.154-0.367)	-62.785 (-78.847 to -31.722)
Female								
Absolute number	162.933 (135.471-194.887)	34.12 (27.013-42.355)	421.695 (337.989-539.386)	60.76 (51.29-70.635)	262.679 (137.565-529.017)	-41.335 (-76.159 to 17.767)	7.738 (4.186-13.534)	-14.577 (-60.96 to 76.262)
Age-standardized rate	4.01 (3.291-4.77)	-12.807 (-15.251 to -10.362)	10.033 (7.977-12.569)	-2.829 (-3.623 to -2.006)	6.675 (3.382-13.708)	-59.902 (-83.701 to -20.954)	0.171 (0.092-0.31)	-56.172 (-79.885 to -13.112)
SDI								
Low SDI								
Absolute number	70.537 (55.929-86.656)	73.363 (63.718-83.154)	174.936 (131.892-221.506)	125.646 (120.462-130.92)	217.504 (100.188-390.428)	-12.189 (-63.016 to 84.85)	3.993 (1.824-7.504)	-2.052 (-59.286 to 108.567)
Age-standardized rate	4.254 (3.498-5.158)	-17.355 (-20.073 to -14.503)	11.343 (9.031-14.587)	-1.018 (-1.248 to -0.815)	13.058 (6.206-23.763)	-55.027 (-81.008 to -7.249)	0.321 (0.146-0.61)	-53.471 (-81.143 to 3.368)
Low-middle SDI								
Absolute number	47.454 (37.998-58.792)	50.548 (42.875-58.68)	136.883 (105.679-174.533)	79.665 (71.117-86.843)	108.412 (55.005-197.632)	-42.346 (-74.408 to 25.921)	2.333 (1.121-4.459)	-25.424 (-66.705 to 64.21)
Age-standardized rate	4.1 (3.376-5.015)	-9.902 (-12.328 to -7.729)	11.776 (9.316-14.998)	-1.24 (-1.523 to -1.017)	9.986 (5.036-18.362)	-59.781 (-81.79 to -10.421)	0.25 (0.119-0.492)	-56.615 (-81.182 to -0.74)
Middle SDI								
Absolute number	39.416 (31.605-47.933)	44.508 (35.668-54.499)	117.959 (91.427-150.7)	67.266 (54.353-79.433)	63.836 (37.504-92.553)	-49.134 (-68.284 to -17.935)	1.798 (0.94-2.594)	-21.023 (-51.389 to 34.513)
Age-standardized rate	4.165 (3.383-5.043)	-11.498 (-13.889 to -9.33)	12.114 (9.502-15.225)	-3.268 (-3.912 to -2.595)	7.377 (4.337-10.745)	-64.86 (-77.909 to -43.154)	0.236 (0.12-0.344)	-57.695 (-75.31 to -28.954)
High-middle SDI								
Absolute number	82.202 (67.293-99.083)	26.397 (17.778-37.177)	223.682 (175.917-289.759)	49.318 (34.697-65.367)	118.448 (68.281-163.521)	-63.981 (-75.949 to -41.973)	3.87 (2.057-5.633)	-40.674 (-62.883 to -1.531)
Age-standardized rate	5.032 (4.114-6.01)	-14.558 (-16.98 to -12.024)	12.885 (10.127-16.273)	-5.294 (-6.29 to -4.279)	7.894 (4.663-10.818)	-74.587 (-83.375 to -58.392)	0.228 (0.122-0.328)	-72.543 (-82.915 to -53.057)
High SDI								
Absolute number	149.703 (126.048-174.362)	15.034 (9.894-21.131)	381.986 (309.794-488.175)	28.607 (19.206-39.728)	140.649 (114.997-165.836)	-55.233 (-66.818 to -38.936)	4.956 (3.903-6.111)	-28.41 (-47.151 to -1.352)
Age-standardized rate	6.304 (5.151-7.482)	-2.88 (-5.449 to -0.086)	13.75 (10.904-17.118)	-1.474 (-3.574 to 0.7)	5.308 (4.38-6.204)	-67.707 (-75.814 to -56.696)	0.133 (0.106-0.162)	-62.595 (-72.128 to -48.688)

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TABLE 1 Continued

	Prevalence (95% UI)		Incidence (95% UI)		DALYs (95% UI)		Deaths (95% UI)	
	2023	Percent change, 1990-2023	2023	Percent change, 1990-2023	2023	Percent change, 1990-2023	2023	Percent change, 1990-2023
GBD regions								
Central Asia								
Absolute number	4.081 (3.244-4.991)	28.812 (20.811-35.813)	12.011 (9.187-14.97)	48.291 (39.876-56.482)	9.919 (7.63-12.535)	-9.356 (-36.067 to 33.846)	0.232 (0.176-0.303)	-0.69 (-31.785 to 47.461)
Age-standardized rate	4.108 (3.278-5.04)	-6.692 (-9.775--4.17)	12.118 (9.361-15.279)	1.098 (0.637-1.626)	10.13 (7.777-12.901)	-38.698 (-56.814 to -9.489)	0.263 (0.199-0.348)	-41.972 (-60.408 to -14.634)
Central Europe								
Absolute number	6.067 (5.119-7.105)	-21.369 (-26.357 to -16.089)	13.921 (11.076-17.975)	-3.255 (-9.87 to 5.158)	9.796 (8.315-11.46)	-61.197 (-67.608 to -51.751)	0.408 (0.341-0.494)	-38.001 (-49.086 to -22.178)
Age-standardized rate	5.862 (4.734-7.022)	-10.085 (-14.483 to -6.032)	11.193 (8.674-14.363)	0.853 (0.546-1.108)	7.063 (6.022-8.167)	-67.613 (-72.336 to -60.394)	0.215 (0.181-0.258)	-60.766 (-67.031 to -51.209)
Eastern Europe								
Absolute number	8.763 (7.167-10.667)	-28.438 (-34.217 to -22.531)	27.666 (21.854-36.241)	-6.933 (-13.242 to -0.765)	23.859 (18.132-30.253)	6.192 (-26.128 to 56.74)	0.815 (0.636-1.015)	9.099 (-23.568 to 61.573)
Age-standardized rate	4.509 (3.618-5.539)	-21.495 (-26.726 to -16.726)	12.792 (9.946-16.33)	-0.038 (-0.389 to 0.215)	9.553 (7.304-12.088)	-10.641 (-37.891 to 29.357)	0.277 (0.216-0.347)	-25.251 (-47.804 to 12.383)
High-income Asia Pacific								
Absolute number	12.724 (10.693-14.864)	29.751 (22.025-40.45)	20.917 (16.604-26.823)	14.864 (2.818-33.266)	11.476 (9.392-13.653)	-38.293 (-54.861 to -14.716)	0.43 (0.324-0.536)	13.072 (-18.803 to 61.046)
Age-standardized rate	9.807 (7.822-12.24)	55.346 (43.513-68.813)	10.051 (7.718-12.907)	-0.454 (-0.857 to -0.104)	6.464 (5.346-7.642)	-48.05 (-62.669 to -30.246)	0.127 (0.102-0.153)	-45.977 (-60.254 to -25.336)
Australasia								
Absolute number	1.761 (1.473-2.062)	11.356 (3.821-19.988)	4.27 (3.455-5.361)	68.046 (58.937-81.199)	1.117 (0.945-1.298)	-61.97 (-69.87 to -52.966)	0.026 (0.022-0.031)	-53.694 (-63.817 to -41.281)
Age-standardized rate	5.902 (4.831-6.971)	-28.797 (-33.44 to -24.761)	11.902 (9.284-15.129)	0.062 (-1.157 to 1.625)	3.554 (3.019-4.106)	-76.48 (-81.459 to -70.793)	0.066 (0.056-0.078)	-75.978 (-81.31 to -69.859)
High-income North America								
Absolute number	29.864 (25.325-34.271)	14.712 (8.465-21.432)	75.535 (63.854-90.585)	41.165 (27.789-55.757)	16.66 (13.012-20.132)	-32.643 (-49.799 to -12.02)	0.33 (0.258-0.406)	-20.133 (-41.55 to 5.255)
Age-standardized rate	8.762 (7.209-10.264)	-10.88 (-15.588 to -6.605)	17.354 (14.414-20.886)	-1.361 (-10.168 to 7.58)	5.053 (3.921-6.139)	-46.739 (-60.621 to -30.379)	0.081 (0.063-0.1)	-44.742 (-60.249 to -26.693)
Southern Latin America								
Absolute number	2.061 (1.732-2.426)	13.968 (7.174-22.766)	6.606 (5.435-8.044)	49.038 (40.83-59.646)	1.683 (1.257-2.189)	-68.917 (-79.448 to -54.064)	0.058 (0.043-0.074)	-43.423 (-63.367 to -16.104)
Age-standardized rate	2.841 (2.387-3.385)	-21.948 (-25.939 to -18.387)	8.612 (7.072-10.451)	-5.134 (-9.438 to -0.89)	2.363 (1.757-3.1)	-78.419 (-85.747 to -67.908)	0.065 (0.048-0.085)	-71.381 (-81.815 to -57.298)
Western Europe								
Absolute number	28.393 (24.097-32.853)	22.202 (17.858-28.318)	75.222 (62.502-94.006)	17.316 (11.823-22.946)	15.578 (13.282-17.832)	-41.257 (-51.635 to -30.966)	0.6 (0.477-0.702)	-15.027 (-30.725 to 1.219)
Age-standardized rate	7.064 (5.817-8.341)	12.537 (8.182-17.881)	15.829 (12.474-19.733)	2.7 (-0.887 to 5.827)	3.23 (2.715-3.705)	-56.149 (-63.694 to -48.018)	0.073 (0.061-0.085)	-53.75 (-62.088 to -45.247)

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TABLE 1 Continued

	Prevalence (95% UI)		Incidence (95% UI)		DALYs (95% UI)		Deaths (95% UI)	
	2023	Percent change, 1990-2023	2023	Percent change, 1990-2023	2023	Percent change, 1990-2023	2023	Percent change, 1990-2023
Andean Latin America								
Absolute number	1.968 (1.561-2.511)	76.713 (65.005-88.541)	7.258 (5.721-9.545)	98.649 (83.473-114.389)	1.11 (0.805-1.601)	-36.212 (-64.349 to 11.643)	0.023 (0.015-0.036)	-17.627 (-53.367 to 44.37)
Age-standardized rate	2.951 (2.35-3.74)	-6.933 (-9.361 to -4.975)	10.818 (8.576-14.071)	0.057 (-0.018 to 0.14)	1.785 (1.296-2.575)	-57.756 (-74.867 to -28.748)	0.037 (0.025-0.057)	-59.028 (-76.349 to -30.156)
Caribbean								
Absolute number	1.865 (1.528-2.262)	42.312 (35.948-49.466)	5.469 (4.357-7.199)	50.267 (40.875-60.908)	6.138 (3.191-12.444)	-8.704 (-55.261 to 58.194)	0.1 (0.054-0.203)	4.024 (-47.229 to 73.501)
Age-standardized rate	3.869 (3.169-4.693)	3.688 (1.755-6.482)	10.818 (8.578-14.066)	-0.202 (-0.348 to -0.067)	13.741 (7.064-27.582)	-20.381 (-60.873 to 39.842)	0.213 (0.113-0.431)	-18.458 (-58.322 to 36.519)
Central Latin America								
Absolute number	8.764 (7.035-11.053)	64.596 (53.806-78.024)	30.553 (24.06-39.807)	85.08 (69.036-102.57)	5.615 (4.698-6.524)	-12.243 (-27.902 to 6.717)	0.102 (0.086-0.119)	17.22 (-3.296 to 45.231)
Age-standardized rate	3.333 (2.679-4.141)	-2.42 (-4.062 to -1.123)	11.381 (9.047-14.631)	-0.206 (-0.328 to -0.097)	2.412 (2.01-2.807)	-26.821 (-39.656 to -11.305)	0.042 (0.036-0.049)	-22.476 (-35.684 to -5.003)
Tropical Latin America								
Absolute number	8.975 (7.413-10.967)	54.725 (42.843-66.686)	28.073 (22.221-36.375)	66.99 (50.325-84.814)	8.75 (7.473-10.221)	-30.905 (-42.723 to -15.944)	0.223 (0.185-0.261)	12.572 (-7.492 to 40.313)
Age-standardized rate	4.064 (3.337-4.956)	4.288 (1.605-7.014)	11.928 (9.425-15.252)	-0.151 (-0.218 to -0.079)	4.171 (3.563-4.864)	-49.517 (-58.291 to -38.545)	0.095 (0.078-0.111)	-42.175 (-52.834 to -28.148)
North Africa and Middle East								
Absolute number	21.269 (17.117-26.139)	87.657 (75.224-99.703)	64.46 (49.879-83.402)	112.847 (97.568-128.109)	42.883 (21.632-73.712)	-37.607 (-73.757 to 33.566)	0.945 (0.486-1.65)	-17.466 (-63.156 to 78.305)
Age-standardized rate	3.412 (2.809-4.168)	-3.588 (-5.345 to -2.018)	10.227 (8.051-13.129)	0.011 (-0.398 to 0.499)	7.593 (3.838-13.047)	-59.625 (-81.905 to -18.154)	0.203 (0.104-0.352)	-57.735 (-80.084 to -8.657)
South Asia								
Absolute number	78.803 (63.371-96.933)	56.613 (47.374-66.098)	223.998 (173.104-287.905)	93.303 (83.09-102.737)	202.555 (99.184-363.827)	-35.619 (-72.278 to 56.68)	4.28 (2.016-8.284)	-18.92 (-65.123 to 98.68)
Age-standardized rate	4.261 (3.473-5.174)	-11.076 (-14.175 to -8.523)	11.935 (9.44-15.3)	-0.983 (-1.262 to -0.754)	11.956 (5.847-21.774)	-57.961 (-81.873 to 3.134)	0.281 (0.131-0.544)	-57.644 (-81.042 to 5.898)
East Asia								
Absolute number	98.887 (81.282-117.144)	12.053 (3.766-22.388)	231.712 (182.933-301.377)	22.328 (7.858-38.371)	140.643 (76.753-205.013)	-71.552 (-83.622 to -47)	5.509 (2.845-8.2)	-45.412 (-70.403 to 0.702)
Age-standardized rate	7.041 (5.679-8.539)	-6.384 (-10.419 to -1.758)	15.088 (11.58-19.299)	-1.342 (-1.996 to -0.771)	9.915 (5.491-14.349)	-77.93 (-87.126 to -58.295)	0.296 (0.154-0.435)	-75.586 (-86.86 to -53.196)
Oceania								
Absolute number	0.546 (0.417-0.71)	127.438 (120.738-135.115)	1.908 (1.429-2.49)	130.445 (123.669-137.24)	0.91 (0.437-1.649)	56.237 (-30.341 to 213.138)	0.014 (0.006-0.025)	57.985 (-29.192 to 207.454)
Age-standardized rate	3.823 (2.987-4.877)	-0.152 (-1.719 to 1.9)	13.61 (10.492-17.613)	-0.008 (-0.095 to 0.077)	5.806 (2.764-10.424)	-27.866 (-67.265 to 35.52)	0.1 (0.047-0.18)	-33.092 (-69.757 to 27.023)
Southeast Asia								
Absolute number	25.918 (20.587-32.4)	41.983 (32.327-52.567)	89.479 (69.812-113.338)	67.313 (53.471-80.769)	30.304 (15.568-55.294)	-21.62 (-61.647 to 58.008)	0.894 (0.429-1.636)	15.269 (-46.53 to 147.324)

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TABLE 1 Continued

	Prevalence (95% UI)		Incidence (95% UI)		DALYs (95% UI)		Deaths (95% UI)	
	2023	Percent change, 1990-2023	2023	Percent change, 1990-2023	2023	Percent change, 1990-2023	2023	Percent change, 1990-2023
Age-standardized rate	3.64 (2.935-4.511)	-12.384 (-15.549 to -9.725)	12.214 (9.545-15.176)	0.073 (-0.526 to 0.819)	4.75 (2.419-8.752)	-48.006 (-75.448 to 7.236)	0.147 (0.071-0.272)	-51.188 (-77.661 to 6.151)
Central Sub-Saharan Africa								
Absolute number	6.318 (4.96-7.809)	168.291 (156.176-181.543)	13.485 (10.014-17.631)	174.236 (170.991-178.274)	30.576 (9.226-74.711)	39.589 (-58.703 to 194.726)	0.525 (0.147-1.345)	52.633 (-57.835 to 213.252)
Age-standardized rate	4.49 (3.648-5.391)	1.653 (-1.282 to 4.883)	10.792 (8.545-14.064)	0.003 (-0.18 to 0.237)	22.154 (6.445-56.305)	-41.531 (-82.935 to 19.379)	0.494 (0.131-1.477)	-37.456 (-82.355 to 30.197)
Eastern Sub-Saharan Africa								
Absolute number	22.547 (17.84-27.161)	46.415 (35.881-59.438)	43.693 (32.34-56.445)	149.955 (144.846-154.952)	36.827 (15.91-70.947)	-9.892 (-67.003 to 93.338)	0.558 (0.223-1.139)	-8.074 (-68.762 to 105.069)
Age-standardized rate	5.118 (4.203-6.018)	-33.548 (-37.27 to -29.668)	11.25 (8.955-14.507)	-0.086 (-0.22 to 0.026)	7.796 (3.394-15.441)	-60.01 (-84.989 to -15.017)	0.147 (0.06-0.314)	-60.643 (-85.567 to -15.99)
Southern Sub-Saharan Africa								
Absolute number	3.305 (2.595-4.115)	75.559 (67.412-84.991)	10.142 (7.764-13.003)	88.721 (77.41-99.215)	6.564 (3.328-10.591)	28.658 (-37.791 to 163.392)	0.12 (0.059-0.199)	38.242 (-31.016 to 182.7)
Age-standardized rate	3.772 (3.032-4.66)	-0.751 (-2.351 to 0.912)	11.623 (9.204-14.907)	0.277 (0.177-0.376)	7.746 (3.94-12.499)	-17.627 (-58.507 to 69.6)	0.157 (0.077-0.263)	-25.923 (-61.44 to 48.588)
Western Sub-Saharan Africa								
Absolute number	16.703 (12.753-21.124)	116.169 (102.045-129.572)	49.813 (36.967-64.266)	171.897 (167.221-175.373)	46.659 (20.268-85.191)	-16.323 (-70.961 to 98.772)	0.752 (0.318-1.418)	-16.622 (-71.742 to 97.461)
Age-standardized rate	3.536 (2.854-4.432)	-16.589 (-20.296 to -13.354)	11.317 (9.013-14.565)	-0.992 (-1.368 to -0.683)	8.63 (3.817-16.127)	-67.434 (-88.286 to -23.761)	0.195 (0.083-0.366)	-68.006 (-89.026 to -26.117)

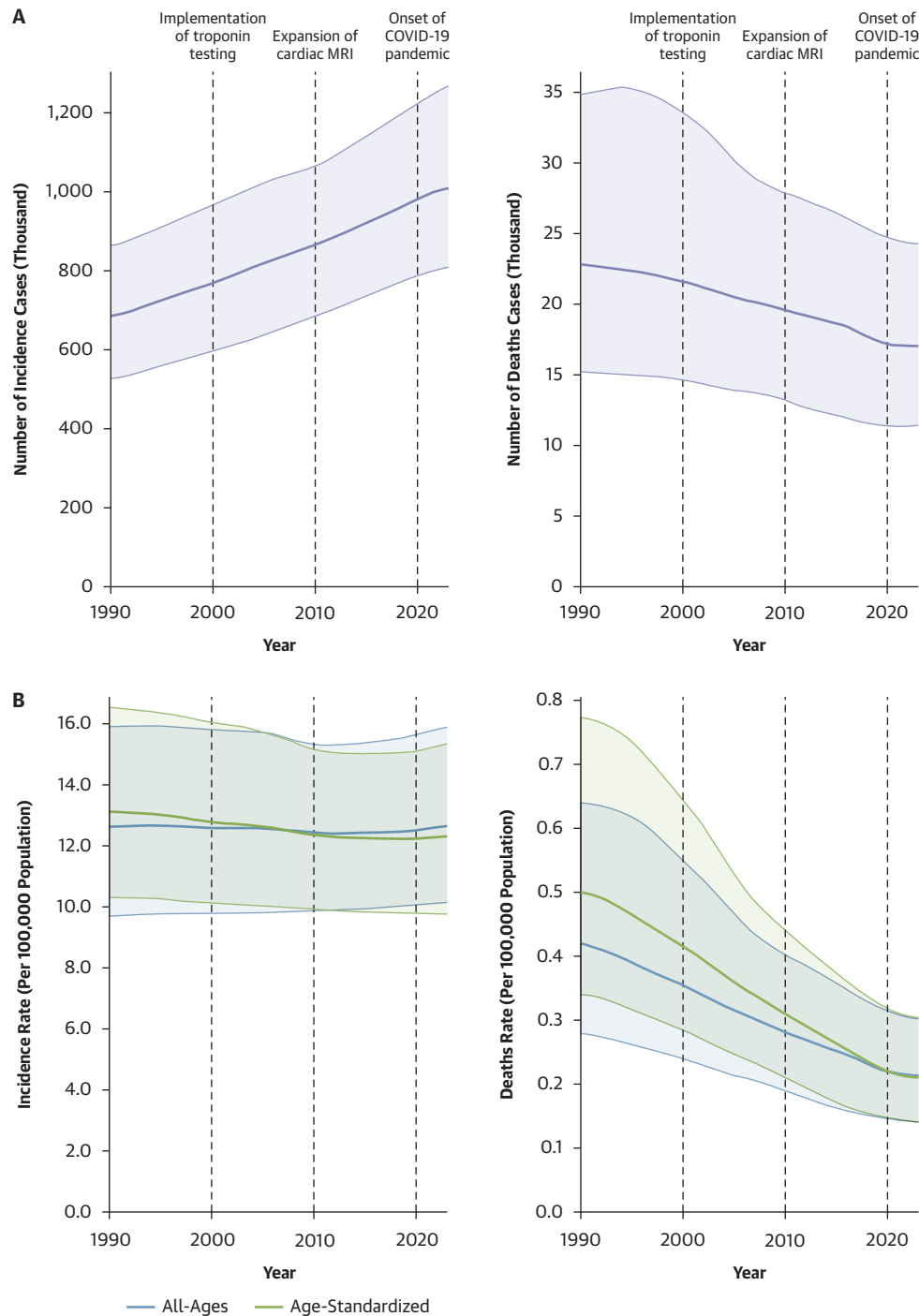
DALY = disability-adjusted life-year; GBD = Global Burden of Disease, Injuries, and Risk Factors Study; SDI = Socio-demographic Index; UI = uncertainty interval.

10.90-17.12] in 2023; high-middle SDI: 12.50 [95% UI: 9.89-15.47] in 2020 and 12.89 [95% UI: 10.13-16.27] in 2023; middle SDI: 11.91 [95% UI: 9.37-14.87] in 2020 and 12.11 [95% UI: 9.50-15.23] in 2023), whereas low-middle and low SDI countries remained stable throughout this period. Detailed results for each estimate are presented in [Supplemental Tables 5 to 8](#). The decomposition of DALY numbers also differed substantially by SDI level ([Supplemental Table 9](#)). Globally, total DALY numbers decreased by 555.80 (95% UI: -860.86 to -306.29) thousand from 1990 to 2023, driven predominantly by epidemiologic change (-931.05 [95% UI: -1256.11 to -665.13] thousand), while population growth partially offset this reduction (413.57 thousand [95% UI: 358.15-482.25 thousand]). High SDI countries followed a similar pattern, with a greater reduction attributable to epidemiologic change and a modest increase attributable to

population growth. In contrast, lower SDI countries showed attenuated overall reductions, as the DALY burden contributed by population growth largely offset the reductions attained through epidemiologic change.

BURDEN OF HEART FAILURE DUE TO MYOCARDITIS. The number of prevalent cases of heart failure attributable to myocarditis remained stable, with 130,028.98 cases (95% UI: 104,069.43-164,303.31 cases) in 1990 and 133,684.43 cases (95% UI: 106,633.83-171,210.37 cases) in 2023. Similarly, age-standardized prevalence rates per 100,000 population showed a nonsignificant downward trend from 2.33 (95% UI: 1.90-2.90) in 1990 and 1.76 (95% UI: 1.41-2.29) in 2023 ([Figure 5](#), [Supplemental Table 10](#)). Additionally, the prevalence rate of heart failure attributed to myocarditis varied across different age

FIGURE 2 Total Numbers and Rates of Myocarditis at the Global Level, 1990-2023, Highlighting Divergence Between Stable Age-Standardized Incidence Rates and Declining Age-Standardized Death Rates



(A) Total number of incident and deaths cases. (B) Age-standardized incidence and death rates (per 100,000 population). (C) Distribution of mortality-to-incidence ratios of myocarditis by age group, 1990-2023. Shaded regions indicate 95% UIs. MIR is the number of deaths divided by the number of incident cases, presented as a percentage. MIR = mortality-to-incidence ratio; UI = uncertainty interval.

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FIGURE 2 Continued

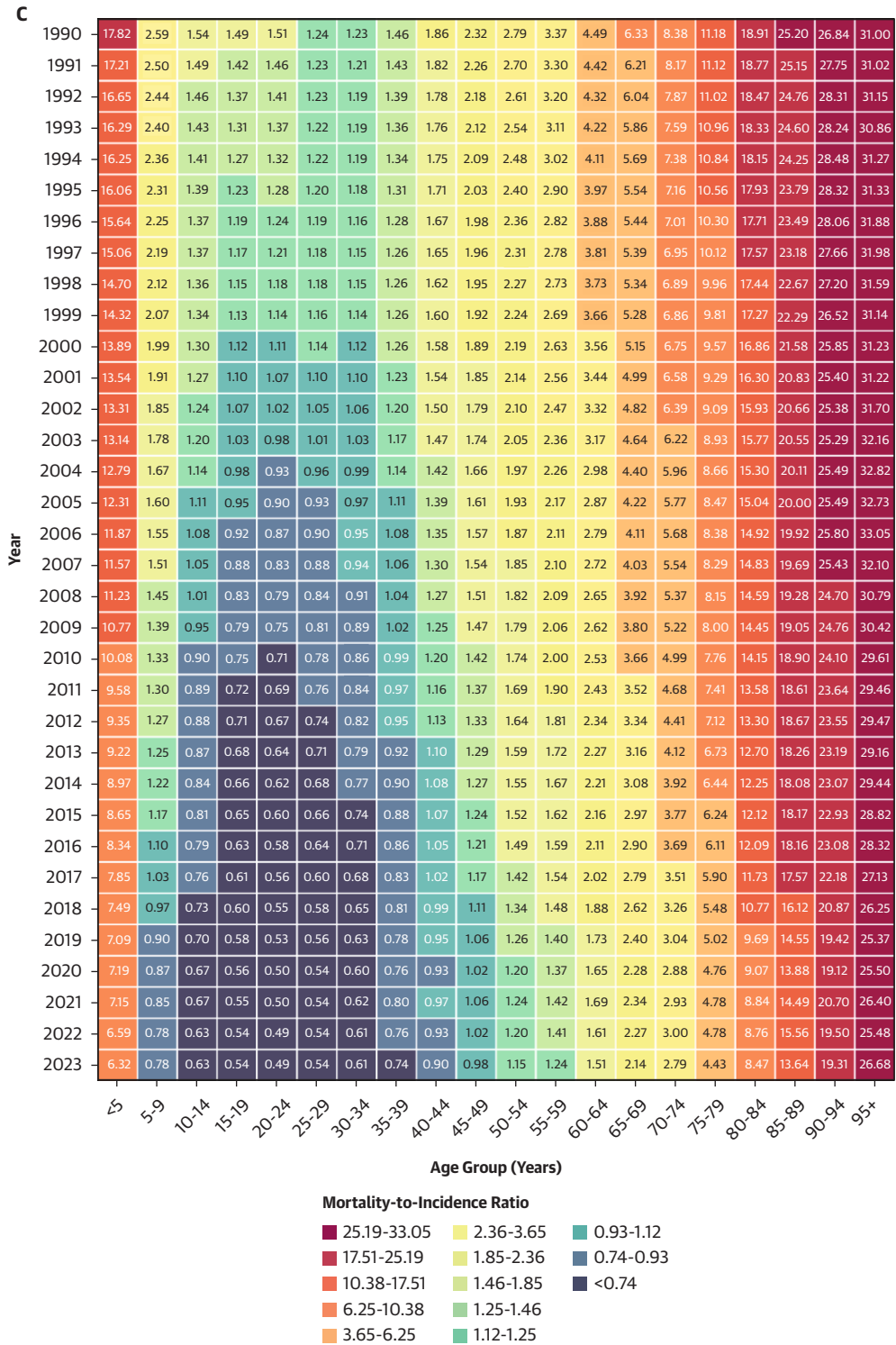
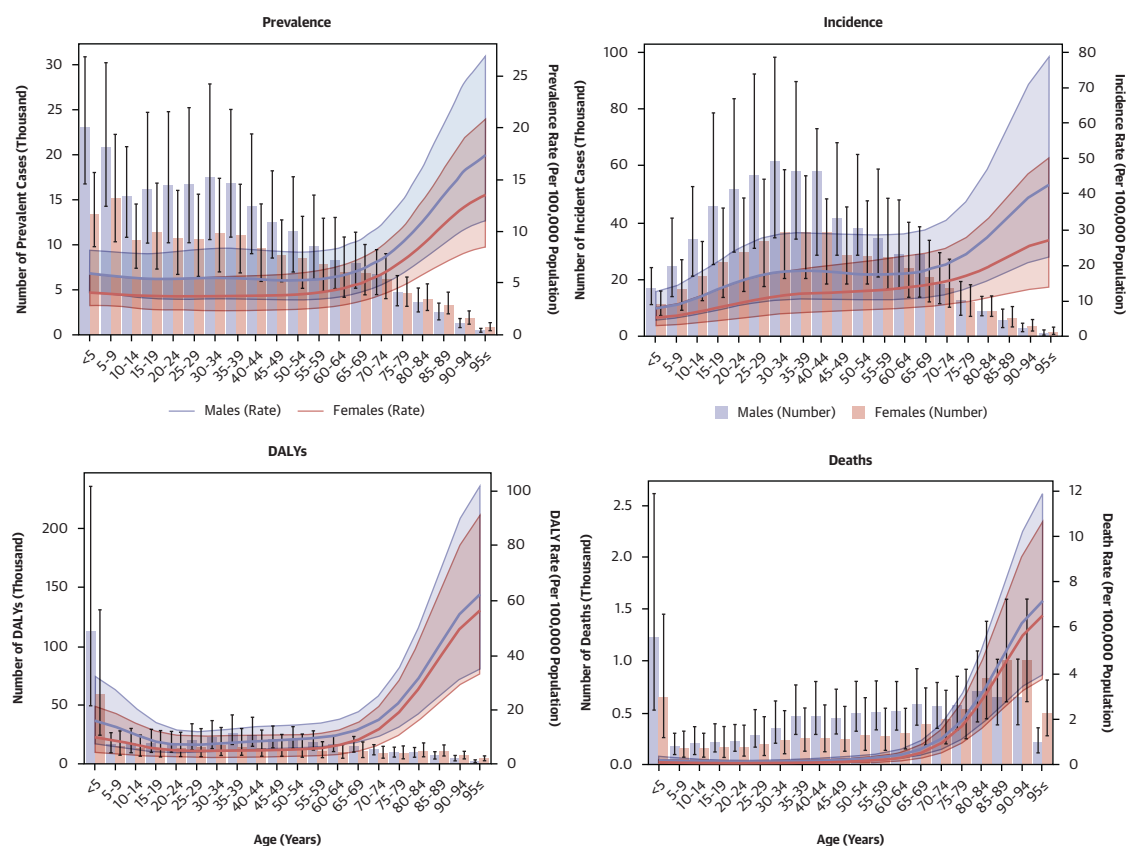


FIGURE 3 Global Age- and Sex-Specific Numbers and Rates (per 100,000 Population) of Prevalence, Incidence, DALYs, and Deaths Due to Myocarditis (2023) With DALYs Showing a U-Shaped Age Pattern



Higher estimates observed in males may be explained by differences in population size. Error bars and shaded regions indicate 95% UIs.
DALY = disability-adjusted life year.

groups, showing a U-shaped curve ([Figure 5](#), [Supplemental Table 11](#)). The group <5 years of age (4.65 [95% UI: 3.25-6.50] per 100,000 population) and 5-9 years of age (3.70 [95% UI: 2.26-5.79]) had the highest prevalence rate among the group <80 years of age, whereas the group between 55-59 years of age had the lowest prevalence rate (0.55 [95% UI: 0.33-0.88]).

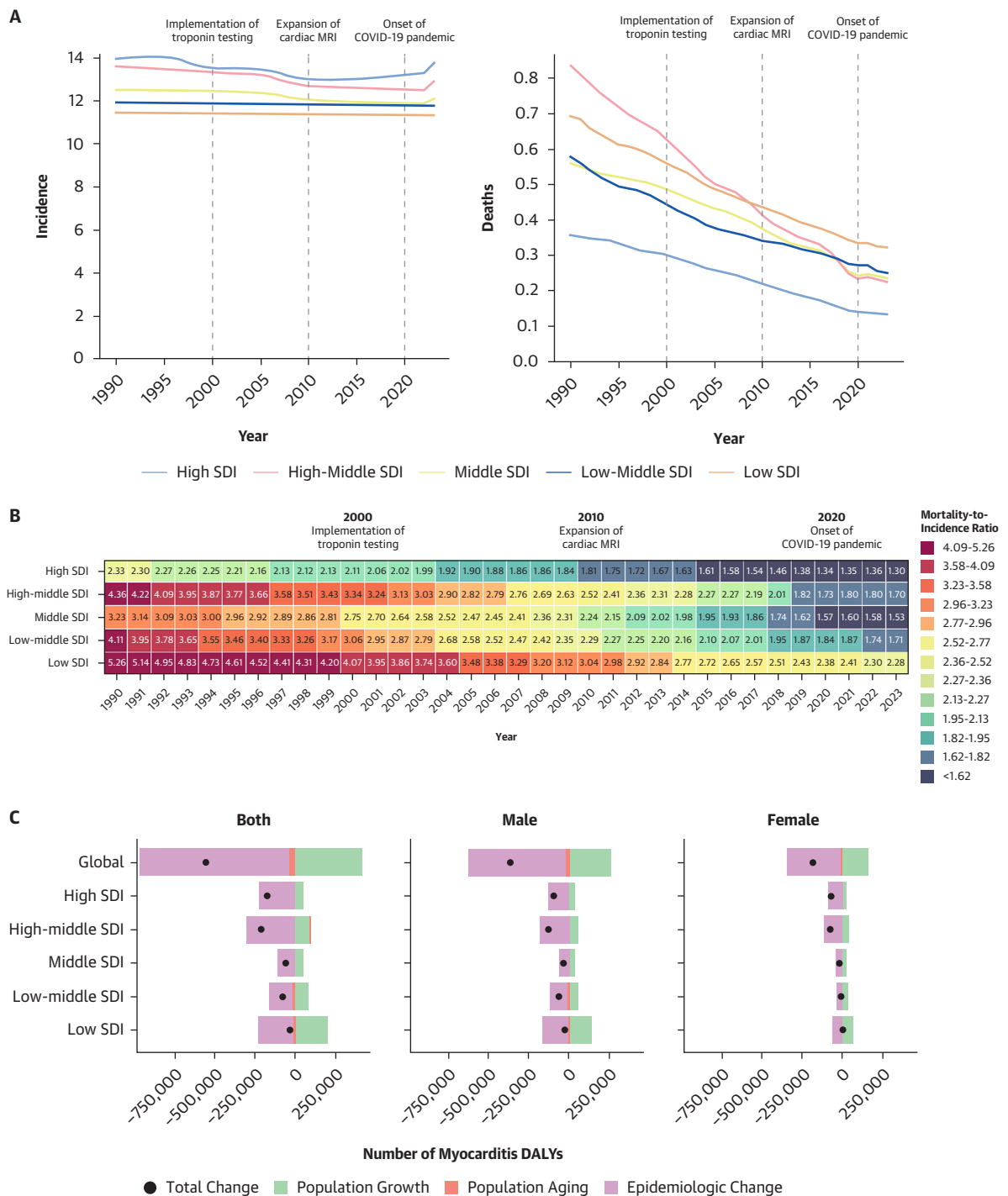
In 2023, the age-standardized prevalence of heart failure attributable to cardiovascular diseases was 523.72 per 100,000 population (95% UI: 468.03-587.02 per 100,000 population), of which myocarditis accounted for a small proportion, at 1.76 per 100,000 population (95% UI: 1.41-2.29 per 100,000 population) ([Figure 5](#), [Supplemental Table 12](#)). However, the relative contribution of myocarditis to heart failure was greater in younger age groups. For example, among children <5 years of age, the prevalence of heart failure due to cardiovascular diseases

was 91.40 per 100,000 population (95% UI: 71.00-117.63 per 100,000 population), of which 4.65 per 100,000 population (95% UI: 3.25-6.50 per 100,000 population) was attributable to myocarditis, accounting for ~5.09% of all cardiovascular disease-related heart failure cases.

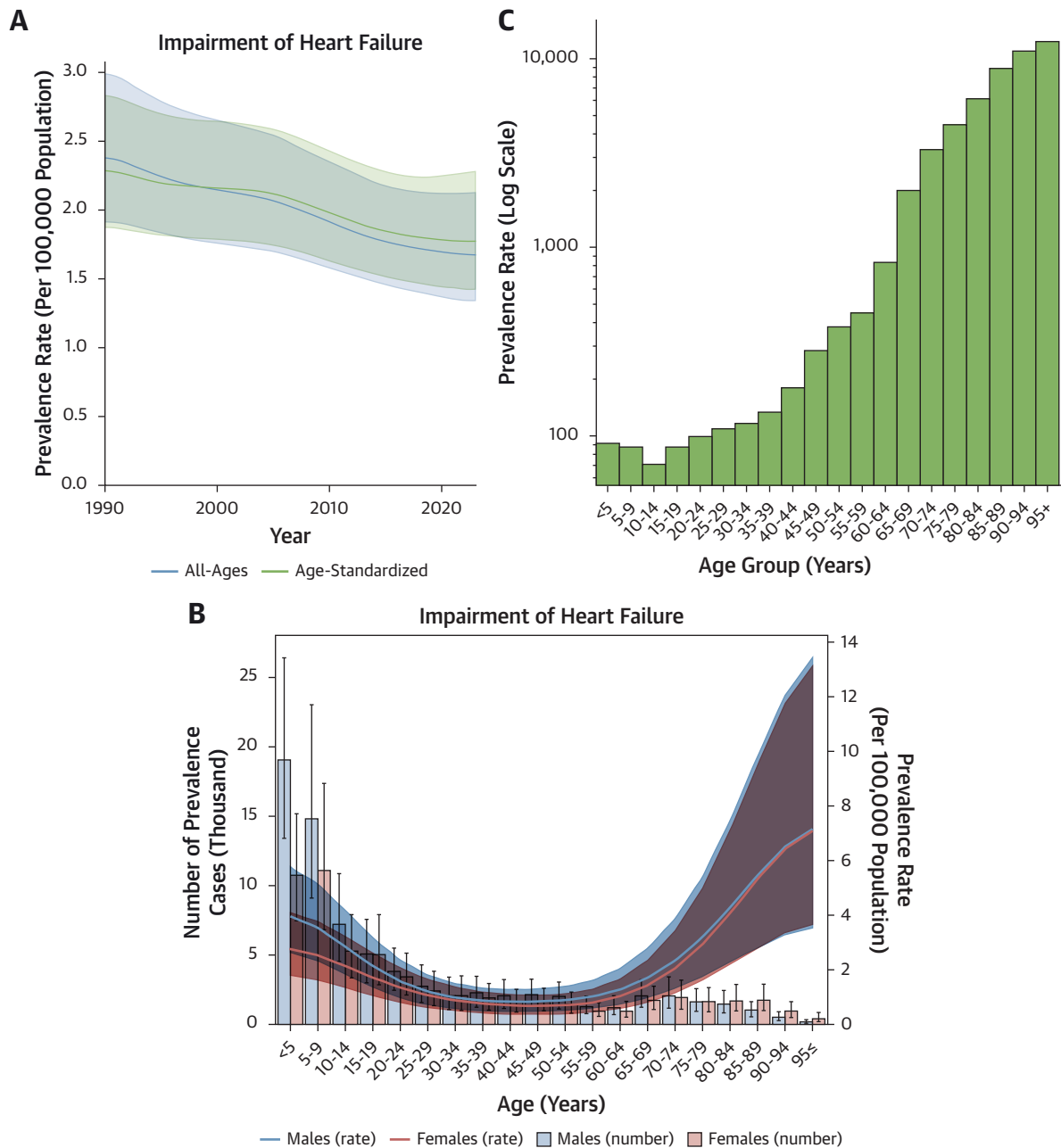
DISCUSSION

KEY FINDINGS. The total number of people with myocarditis has increased from 1990 to 2023, which is likely attributed to population growth and aging, considering that age-standardized prevalence and incidence rates have not increased. Age-standardized estimates of DALY and mortality rates for myocarditis have decreased, which may reflect improvements in early diagnosis and disease management, evolving diagnostic practices, or changes in coding practices. Globally, this reduction in DALY burden

FIGURE 4 Global Trends in Myocarditis Stratified by SDI Groups (1990-2023) Showing Higher Age-Standardized Incidence Rates but Lower Age-Standardized Death Rates in High-SDI Countries



(A) Age-standardized incidence and death rates (per 100,000 population). (B) Distribution of MIRs for myocarditis. (C) Contributions of population growth, population aging, and epidemiological change to changes in DALYs from 1990 to 2023. MIR is the number of deaths divided by the number of diagnosed cases, presented as a percentage. SDI = Socio-demographic Index; other abbreviations as in [Figures 2 and 3](#).

FIGURE 5 Heart Failure Impairment Due to Myocarditis at the Global Level, Showing a U-Shaped Age Pattern in Prevalence Rates

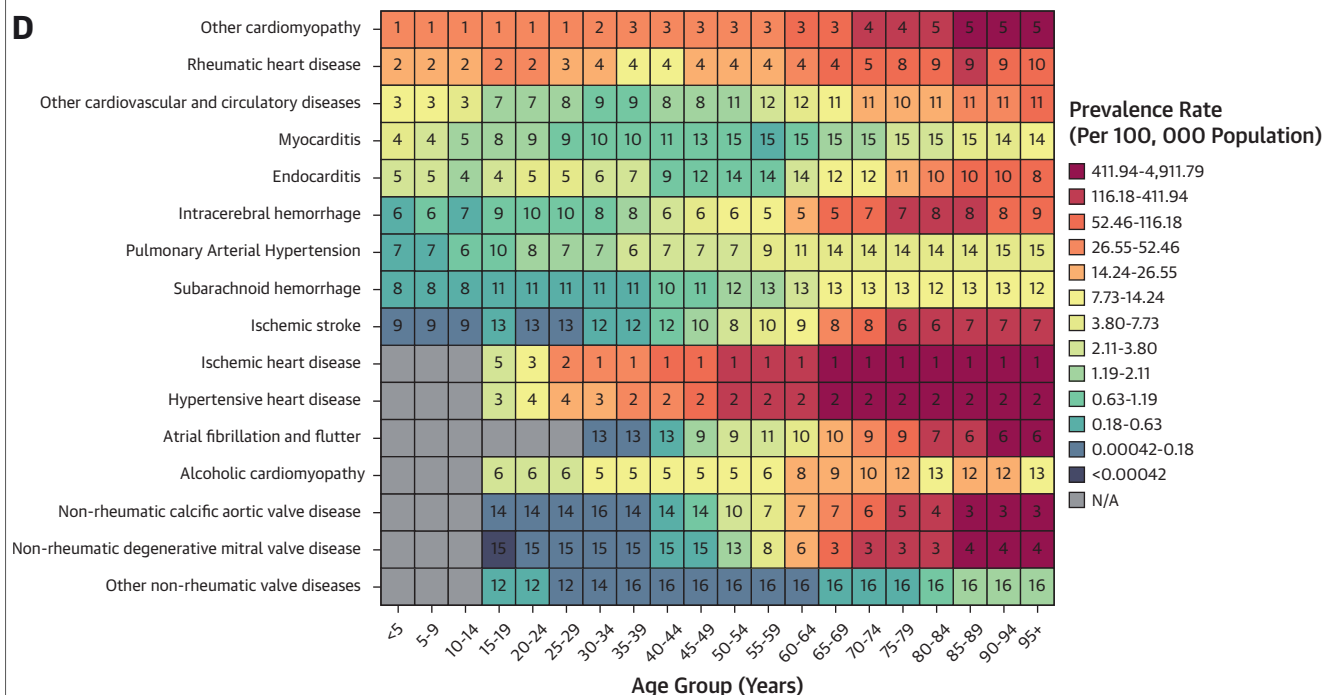
(A) Age-standardized and all-age prevalence rates (per 100,000 population), 1990-2023. (B) Age-specific numbers and rates (per 100,000 population) of prevalence by age group and sex, 2023. (C) Age-specific prevalence rates (per 100,000 population) of heart failure due to cardiovascular diseases in 2023 by age groups and (D) the corresponding rank of each cardiovascular disease. Error bars and shaded regions indicate 95% UIs. Abbreviations as in Figures 2 and 4.

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was driven predominantly by epidemiologic change, though population growth partially offset these gains. Notably, this offsetting effect was more pronounced in lower SDI countries, where population

growth attenuated the reductions attained through epidemiologic change to a greater extent than in higher SDI countries. Males showed higher estimates for every measure, although age-standardized rates

FIGURE 5 Continued

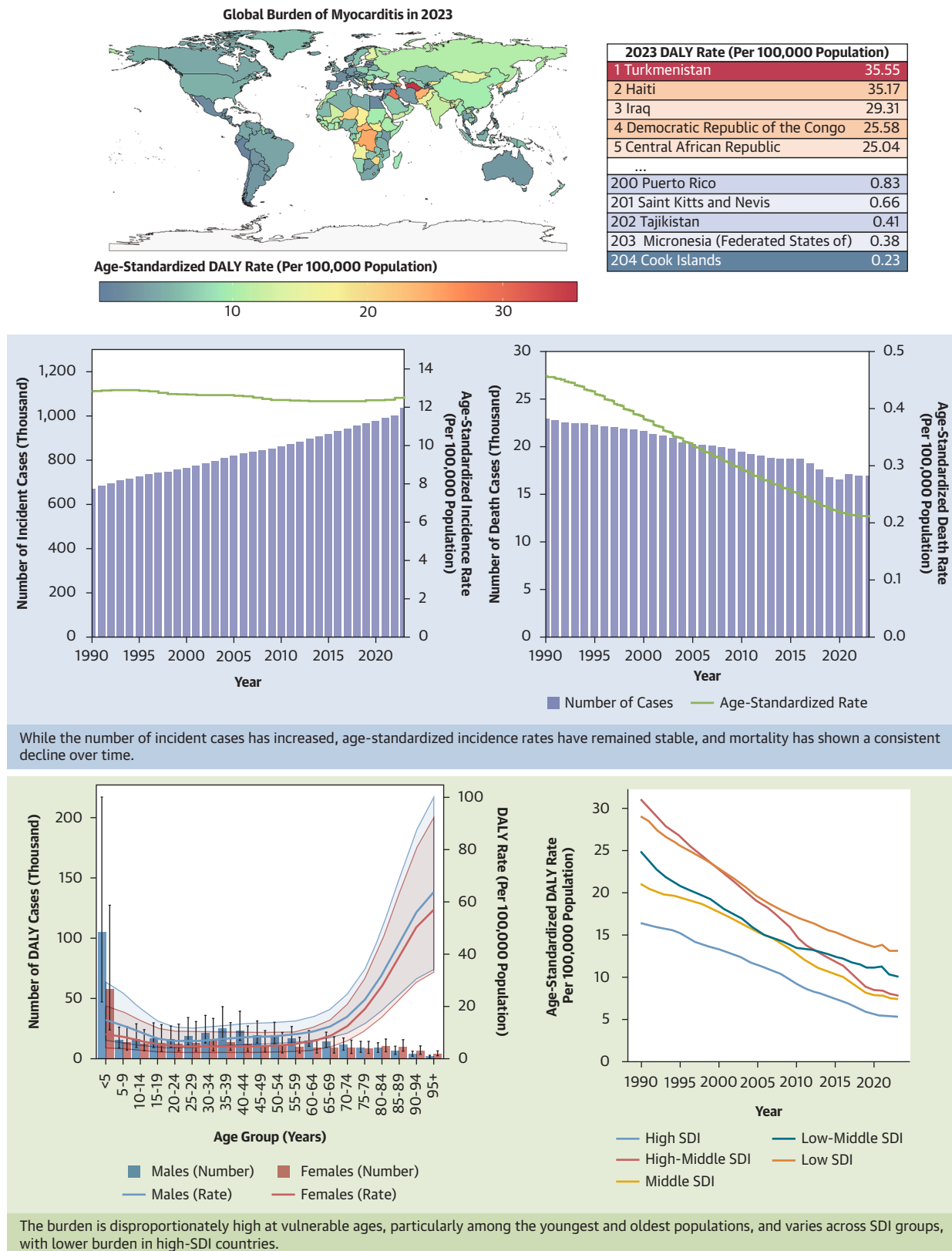


were similar for both sexes, suggesting that observed differences could be attributable to differences in population size and age structure. Moreover, while the age-standardized incidence rates remained steady across all SDI levels, prevalence rates of myocarditis were highest in high SDI countries. Although a post-2020 increase in prevalence and incidence was observed in high and high-middle SDI countries, this trend was not statistically significant. Notably, the youngest and oldest age groups exhibited the highest MIR and the greatest prevalence of heart failure attributable to myocarditis, underscoring their particular vulnerability to severe outcomes.

Although ICD-coded administrative data and death certificates do not provide gold-standard diagnostic certainty, myocarditis can be diagnosed clinically even without advanced imaging or biopsy, and CMR and EMB are increasingly available in many countries. As such, the input data for this analysis represents physicians' diagnostic decisions across a broad range of clinical settings. Administrative surveillance systems offers longitudinal population-level coverage that enables cross-regional and temporal comparisons not otherwise possible. This kind of data forms a foundation for further efforts to improve and extend surveillance of complex cardiac conditions. The GBD 2023 employs consistent,

transparent methods, careful assessment of data quality and comprehensive uncertainty intervals that allow these results to be understood in the context of currently available global information on myocarditis.

INTERPRETATION OF TEMPORAL TRENDS. The observed temporal trends should be interpreted with careful consideration of how changes in clinical recognition, diagnostic intensity, and coding practices may have influenced the reported estimates. Three explanations are particularly relevant. First, the stability of age-standardized incidence and prevalence rates suggests that the underlying population risk of myocarditis may not have substantially changed over time, although this interpretation requires caution given concurrent changes in diagnostic capacity and case ascertainment. Second, periods of heightened clinical awareness may have amplified case identification beyond what technological advances alone would predict. During the COVID-19 pandemic, heightened clinical attention to cardiac symptoms may have increased attribution of other cardiac events to myocarditis,¹⁹ which may partly account for the rise in age-standardized prevalence and incidence rates observed in high and high-middle SDI countries from 2020 onward, where diagnostic resources and clinical vigilance were more

CENTRAL ILLUSTRATION Global Burden of Myocarditis, 1990-2023

Lee S, et al. JACC. 2026;■(■):■-■.

DALY = disability-adjusted life year; SDI = Socio-demographic Index.

readily mobilized. However, disruptions in health care access may have simultaneously limited diagnostic work-up in lower SDI settings.²⁰ Third, temporal changes in ICD coding systems and cause-of-death attribution practices may have independently influenced observed mortality patterns, given the clinical overlap between myocarditis and conditions such as myocardial infarction or dilated cardiomyopathy. Furthermore, the observed inflection points in SDI-stratified temporal trends appear to align with key shifts in diagnostic criteria, including the introduction of cardiac troponin assays in the late 1990s and the expansion of CMR imaging following the Lake Louise criteria around 2009, both of which may have further shaped attribution and classification practices over time. Overall, these patterns most plausibly reflect the combined influence of relatively stable underlying disease occurrence, evolving diagnostic intensity, and changing coding and attribution practices.

COMPARISON WITH THE GBD 2023 CARDIOVASCULAR CAPSTONE ANALYSIS. Our findings are broadly consistent with the GBD 2023 cardiovascular capstone analysis.²¹ The overall increase in myocarditis cases appears to be largely driven by population growth and aging rather than changes in age-standardized incidence, and age-standardized mortality and DALY rates declined over time. However, myocarditis exhibits several epidemiologic features that distinguish it from broader cardiovascular disease patterns. In contrast to other cardiovascular conditions where modifiable risk factors play a substantial role, myocarditis incidence remained relatively stable across time and SDI levels, suggesting a comparatively modest role of traditional risk factors. The burden disproportionately affected the youngest and oldest age groups, and the pattern of higher prevalence but lower mortality in high-SDI settings suggests that myocarditis burden is more strongly influenced by diagnostic practices and case identification than by underlying risk factor profiles.

PLAUSIBLE UNDERLYING MECHANISMS. Compared to 1990, the total number of prevalent, incident, and deaths cases increased in 2023, with no change in adjusted prevalence and incidence rates suggesting that population growth and aging are primary contributors. Several interrelated factors may explain the decline in mortality and DALY rates. In high SDI regions the preferred diagnostic test to confirm acute myocarditis is CMR which has led to increased testing for suspected cases.^{22,23} This shift from EMB to CMR

may have led to an increase in diagnosed cases with less severe presentations. Advances in clinical management may have contributed to improved outcomes, with earlier diagnosis, optimized supportive care, and the wider availability of mechanical circulatory support, such as extracorporeal membrane oxygenation and ventricular assist devices, have improved survival in patients with severe or fulminant myocarditis.^{19,21} Additionally, heterogeneous clinical presentation of myocarditis and its overlap with other cardiac conditions may have contributed to shifts in clinical recognition and cause-of-death attribution over time.²⁴ Broader use of sensitive imaging and biomarkers has likely further increased identification of milder or subclinical presentation.²²

BIOLOGICAL AND ETIOLOGICAL MECHANISMS ACROSS SEX AND AGE GROUPS. Although the absolute numbers of global prevalence, incidence, DALY, and death were slightly higher in males, age-standardized estimates for all ages combined were similar between the sexes. Previous studies have suggested that testosterone may increase viral myocardial damage, while estrogen reduces effector immune responses, affording relative protection.²⁵⁻²⁷ Sex differences in myocarditis rates are generally not observed prior to puberty or after menopause, which aligns with our findings.²⁷

Mortality rates varied across different age groups. Myocarditis resulted in death in ~1 in 15 people (1789.09 deaths per 28,303.21 incident cases) in the group <5 years of age, and 1 in 12 people (1524.42 deaths per 17,998.44 incident cases) in the group between 80-84 years of age. In contrast, the mortality rate was much lower, ~1 in 189 people (390.15 deaths per 71,854.50 incident cases), in the group between 15-19 years of age. This suggests that both the youngest and oldest age groups are at a higher risk of experiencing fatal outcomes.

These differences may be partially explained by immune immaturity during infancy and immune senescence in older age.²⁸ However, severity of myocarditis is also influenced by heterogeneous immunopathological mechanisms, including variation in dominant etiology, adequacy of viral clearance, and the degree of immune-mediated myocardial injury. Young children <5 years of age are more susceptible to virus-induced myocarditis,^{4,29} and impaired viral clearance may prolong viral replication and amplify downstream inflammatory responses, increasing risk of progression to dilated cardiomyopathy following acute myocarditis.³⁰ In older adults, causes are more varied, including

autoimmune diseases,³¹ drug-induced reactions,³² particularly by immune checkpoint inhibitors, which has been reported to carry major adverse cardiovascular event rates of 25% to 50%,³³ and substance use, including cocaine³⁴ and amphetamine, which cause myocardial inflammation through direct myocardial injury.³⁵

Across age groups, exaggerated immune activation, including cytokine-mediated cardiomyocyte injury and T-cell infiltration, may further contribute to myocardial damage beyond the initial pathogenic process.¹ Together, these findings suggest that age-related variation in myocarditis outcomes reflects a combination of host immune factors, etiology-specific disease mechanisms, and the degree of immune dysregulation at the time of myocardial injury.

IMPACT OF THE COVID-19 PANDEMIC ON MYOCARDITIS BURDEN. Population level data on myocarditis collected during years of the COVID-19 pandemic remains limited to national and subnational level datasets from only a limited subset of countries. The association between severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection and myocarditis has been reported, particularly in unvaccinated populations,⁶ although the occurrence of myocarditis itself is extremely low in previously healthy infected individuals.⁶ Any potential increases in myocarditis due to SARS-CoV-2 may have been partially offset by decreased transmission of other cardiotropic viruses, such as enteroviruses, due to social distancing measures.³⁶ However, these findings should be interpreted with caution, as shifts in health care utilization during the pandemic, combined with inconsistencies in clinical documentation and etiological attribution across settings, may have influenced the estimation of myocarditis burden.²⁰

CLINICAL AND POLICY IMPLICATIONS. Although age-standardized incidence rates remain consistent across SDI levels, age-standardized DALY and mortality rates were lowest in high SDI countries, revealing important disparities in disease outcomes that likely reflect differences in diagnostic capacity, clinical recognition, and management rather than underlying disease risk alone. Within the broader cardiovascular disease landscape, myocarditis is distinctive in disproportionately affecting younger individuals and exhibits a high MIRs in the youngest and oldest groups. In these populations, physicians should maintain a lower threshold for diagnostic evaluation and early escalation in these age groups.²²

In high-SDI settings, wider access to advanced imaging and longitudinal care may facilitate the detection of less severe cases and improved survival,

which could contribute to the observed pattern of higher prevalence but lower mortality. However, disparities in access to CMR persist even within these settings, including among rural populations and historically underserved populations, and may influence both case identification and outcomes.³⁷ Additionally, neuropsychiatric comorbidities such as depression and anxiety contribute substantially to nonfatal burden, although their recognition and integration into routine cardiac care remain variable.^{1,38} Equitable access to CMR and the integration of mental health services into cardiac care pathways warrant particular attention in these settings.

In low SDI settings, limited diagnostic capacity may result in under-detection of milder cases, higher likelihood of severe presentations being captured, and greater uncertainty in mortality patterns. Improving diagnostic consistency, access to advanced imaging, and clinical documentation quality would enhance both clinical interpretation and the comparability of burden estimates across settings. These findings highlight the importance of age- and setting-specific approaches to myocarditis diagnosis and management.³⁹

STUDY LIMITATIONS. First, diagnosing myocarditis poses a significant challenge due to its heterogeneous clinical presentations and the need for diagnostic technologies. In addition to a classical clinical presentation with chest pain, electrocardiogram changes and elevated cardiac biomarkers, advanced imaging usually with CMR or tissue analysis from EMB are required to confirm myocarditis.²² Our study relied on clinical diagnosis or postmortem diagnosis made by physicians directly responsible for each patient. Second, underreporting bias exists due to limited diagnostic resources for myocarditis in many countries.⁴⁰ The broad clinical overlap between myocarditis and conditions such as myocardial infarction, genetic cardiomyopathies, and takotsubo cardiomyopathy may result in misclassification, leading to underestimation in low-SDI settings^{24,41} and potential over detection of milder cases in high-SDI settings.⁴² Third, asymptomatic or subclinical presentations may go undetected, including cases occurring silently in conjunction with systemic diseases.⁴³ Fourth, MIR should be interpreted with caution given its sensitivity to the accuracy of incidence estimation; in settings with limited diagnostic capacity, underestimated incidence may inflate MIR values, and regional variation may reflect differences in data quality as much as true prognostic differences. Fifth, data availability was limited in certain regions and pandemic years, with larger uncertainty

intervals in data-sparse locations. Changes in health care-seeking behavior during COVID-19 may have introduced detection bias,²⁰ and cardiovascular manifestations of long COVID⁴⁴ may not be consistently captured as myocarditis in clinical records. Sixth, pre- and post-pandemic subgroup analyses were not feasible within the GBD modeling framework, and separating myocarditis attributable to SARS-CoV-2 infection from that associated with mRNA vaccination remains methodologically challenging. Seventh, inpatient-based incidence data with correction factors may modestly overestimate burden by preferentially capturing severe cases.¹⁶ Eighth, the estimation of heart failure attributable to myocarditis is subject to limitations inherent to a model-driven attribution framework, including limited formal validation, the heterogeneous etiology of heart failure, and reliance on limited geographic data sources for excess mortality estimation. Lastly, mortality estimates are affected by redistribution of deaths assigned to garbage codes, including nonspecific cardiovascular conditions, which varies across countries and over time and may influence comparability of mortality trends.

CONCLUSIONS

The absolute number of myocarditis cases increased from 1990 to 2023, whereas age-standardized incidence and prevalence remained stable. Age-standardized DALY and mortality rates declined substantially over this period. Although incidence was similar across SDI levels, high SDI countries showed higher prevalence but lower mortality, reflecting disparities in diagnostic capacity and access to care. The youngest and oldest age groups exhibited the highest MIR (**Central Illustration**), underscoring their particular vulnerability to severe outcomes. Despite limitations in ICD-based case ascertainment, these population-level findings highlight the need for equitable access to diagnostic technologies, consistent monitoring protocols, and evidence-based treatments to reduce the global burden of myocarditis.

DATA AVAILABILITY To download the data used in these analyses, please visit the Global Health Data Exchange GBD 2023 website at <https://ghdx.healthdata.org/gbd-2023>.

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GBD 2023 MYOCARDITIS COLLABORATORS.

Collaborators. Sooji Lee, MD,¹ Hayeon Lee, PhD,² Min Seo Kim, MD,^{3,4} Elaheh Malakan Rad, MD,⁵ Catherine O. Johnson, PhD,⁶ Laura Lara-Castor, PhD,^{6,7} Hasan Aalruz, PhD,⁸ Yohannes Habtegiorgis Abate, MSc,⁹ Nasir Abbas, PhD,¹⁰ Hedayat Abbastabar, PhD,¹¹ Deldar Morad Abdulah, MPH,¹² Olugbenga Olusola Abiodun, FWACP,¹³ Richard Gyan Aboagye, MPH,^{14,15} Eman Abu-Gharbieh, PhD,^{16,17} Salahdein Aburuz, PhD,^{18,19} Juliana Bunmi Adetunji, PhD,²⁰ Prince Owusu Adoma, PhD,²¹ Leticia Akua Adzigbli, BSc,²² Danish Ahmad, PhD,^{23,24} Firdos Ahmad, PhD,²⁵ Ayman Ahmed, PhD,^{26,27} Haroon Ahmed, PhD,²⁸ Sindew Mahmud Ahmed, MSc,^{29,30} Syed Anees Ahmed, PhD,³¹ Salah Al Awaidey, MSc,^{32,33} Samer O Alalalmeh, BPharm,³⁴ Khalifah A. Aldawsari, MD,^{35,36} Fadwa Naji Alhalaqa, PhD,³⁷ Mohammed Ahmed Al-Kaif, PhD,³⁸ Mahmoud A Alomari, PhD,^{39,40} Mohammed A. Alsabri, MD,^{41,42} Najim Z. Alshahrani, MD,⁴³ Awais Altaf, PhD,^{44,45} Jaffar A. Al-Tawfiq, MD,^{46,47} Safwat Aly, MD,^{48,49} Enrico Ammirati, MD,^{50,51} Hubert Amu, PhD,⁵² Ganiyu Adeniyi Amusa, MD,^{53,54} Catalina Liliana Andrei, PhD,⁵⁵ Anayochukwu Edward Anyasodor, PhD,⁵⁶ Jalal Arabloo, PhD,⁵⁷ Aleksandr Y. Aravkin, PhD,^{6,7,58} Demelash Areda, PhD,⁵⁹ Tahira Ashraf, PhD,⁶⁰ Maha Moh'd Wahbi Atout, PhD,⁶¹ Avinash Aujayeb, MBBS,⁶² Shahkaar Aziz, MS,⁶³ Ahmed Y. Azzam, MD,^{64,65} Domenico Azzolino, PhD,⁶⁶ Ovidiu Constantin Baltatu, PhD,⁶⁷ Mohammad-Mahdi Bastan, MD,^{68,69} Kavita Batra, PhD,⁷⁰ Sonu Bhaskar, MD,^{71,72} Jasvinder Singh Bhatti, PhD,⁷³ Raffaele Bugiardini, MD,⁷⁴ Yasser Bustanji, PhD,^{75,76} Ismael Campos-Nonato, PhD,⁷⁷ Edina Cenko, MD,⁷⁴ Sandip Chakraborty, MVSc,⁷⁸ Jeffrey Shi Kai Chan, MPH,^{79,80} Vijay Kumar Chattu, PhD,^{81,82} An-Tian Chen, PhD,^{83,84} Nicholas W.S. Chew, MD,⁸⁵ Gerald Chi, MD,⁸⁶ Fatemeh Chichagi, MD,^{87,88} Patrick R. Ching, MD,⁸⁹ Bryan Chong, MBBS,⁹⁰ Dinh-Toi Chu, PhD,^{91,92} Natalia Cruz-Martins, PhD,^{93,94} Nour Dababo, MD,^{95,96} Ali Dabbagh, MD,⁹⁷ Omid Dadras, PhD,⁹⁸ Xiaochen Dai, PhD,^{6,7} Mayank Dalakoti, MPH,^{99,100} Emily Corbett Desai, MS,⁶ Mitiku Desalegn, MSc,¹⁰¹ Vinoth Gnana Chelaiyan Devanbu, MD,¹⁰² Devananda Devegowda, PhD,¹⁰³ Syed Masudur Rahman Dewan, PhD,^{104,105} Kuldeep Dhama, PhD,¹⁰⁶ Thao Huynh Phuong Do, MD,¹⁰⁷ Ojas Prakashbhai Doshi, MS,¹⁰⁸ Rajkumar Prakashbhai Doshi, MD,^{109,110} Paulo Magno Martins Dourado, PhD,^{111,112} Andre Rodrigues Duraes, PhD,^{113,114} Oyewole Christopher Durojaiye, MPH,¹¹⁵ David Edvardsson, PhD,¹¹⁶ Michael Ekholuenetale, PhD,¹¹⁷ Islam Y. Elgendy, MD,^{118,119} Muhammed Elhadi, MD,^{120,121} Adel B. Elmoselhi, PhD,^{122,123} Chadi

- Eltaha, MD,¹²⁴ Farshid Etaee, MD,¹²⁵ Adeniyi Francis Fagbamigbe, PhD,^{126,127} Ayesha Fahim, PhD,¹²⁸ Jacopo Farina, MD,¹²⁹ Alireza Feizkhah, MD,¹³⁰ Rodrigo Fernandez-Jimenez, PhD,^{131,132} Nuno Ferreira, PhD,¹³³ Florian Fischer, PhD,¹³⁴ Aravind P. Gandhi, MD,¹³⁵ Genanew Kassie Getahun, MPH,¹³⁶ Sama Ghoba, MS,⁶ Laszlo Göbölös, PhD,^{137,138} Mahaveer Golechha, PhD,¹³⁹ Shi-Yang Guan, MD,¹⁴⁰ Avirup Guha, MD,^{141,142} Rahul Gupta, MBBS,¹⁴³ Rajeev Gupta, MD,^{144,145} Reyna Alma Gutiérrez, PhD,¹⁴⁶ Roberth Steven Gutiérrez-Murillo, PhD,¹⁴⁷ Awoke Derbie Habteyohannes, PhD,¹⁴⁸ Najah R. Hadi, PhD,¹⁴⁹ Samer Hamidi, DrPH,¹⁵⁰ Erin B. Hamilton, MPH,⁶ Ahmad Hammoud, PhD,¹⁵¹ Mahgol Sadat Hassan Zadeh Tabatabaei, MD,¹⁵² Rasmus J Havmoeller, PhD,¹⁵³ Simon I. Hay, DSc,^{6,7} Khezar Hayat, MS,^{154,155} Qiang He, MD,¹⁵⁶ Mehdi Hemmati, MD,^{157,158} Yuta Hiraike, PhD,¹⁵⁹ Mai Hoang, MD,¹⁶⁰ Bing-Fang Hwang, PhD,¹⁶¹ Ramzi Ibrahim, MD,¹⁶² Adalia Ikiroma, PhD,¹⁶³ Olayinka Stephen Ilesanmi, PhD,^{164,165} Muhammad Iqhrammullah, PhD,¹⁶⁶ Md Rabiul Islam, PhD,¹⁶⁷ Chinwe Juliana Iwu-Jaja, PhD,^{168,169} Ammar Abdulrahman Jairoun, PhD,¹⁷⁰ Mihajlo Jakovljevic, PhD,^{171,172} Gwang Hun Jeong, MD,^{173,174} Bijay Mukesh Jeswani, MBBS,¹⁷⁵ Charity Ehimwenma Joshua, BSc,¹⁷⁶ Jacek Jerzy Jozwiak, PhD,¹⁷⁷ Vidya Kadashetti, MDS,¹⁷⁸ Ashish Kumar Kakkar, MD,¹⁷⁹ Vineet Kumar Kamal, PhD,^{180,181} Samuel Berchi Kankam, MD,¹⁸² Paschalis Karakasis, MD,¹⁸³ Arman Karimi Behnagh, MD,^{184,185} Mohammad Keykhaei, MD,^{186,187} Maseer Khan, MD,¹⁸⁸ Mohammad Jobair Khan, MPH,^{189,190} Moien A.B. Khan, MSc,^{191,192} Sezer Kisa, PhD,¹⁹³ Kewal Krishan, PhD,¹⁹⁴ Mukhtar Kulimbet, MSc,^{195,196} Ashish Kumar, MD,¹⁹⁷ Dewesh Kumar, MD,¹⁹⁸ Ville Kytö, MD,^{199,200} Chandrakant Lahariya, MD,^{201,202} Qing Lan, PhD,²⁰³ Seung Won Lee, MD,²⁰⁴ Vasileios Leivaditis, PhD,^{205,206} Ming-Chieh Li, PhD,²⁰⁷ Daniel Lindholm, MD,²⁰⁸ Haipeng Liu, PhD,^{209,210} Alireza Mafi, PhD,^{211,212} Yosef Manla, MD,²¹³ Soroush Masrouri, MD,²¹⁴ Steven M. McPhail, PhD,^{215,216} Elahe Meftah, MD,^{217,218} Tesfahun Mekene Meto, MPH,²¹⁹ Ritesh G. Menezes, MD,²²⁰ George A. Mensah, MD,^{221,222} Tomislav Mestrovic, PhD,^{6,223} Ted R. Miller, PhD,²²⁴ Erkin M. Mirrakhimov, MD,^{225,226} Yidnek Mogessie, MD,^{227,228} Mona Gamal Mohamed, PhD,^{229,230} Abdollah Mohammadian-Hafshejani, PhD,²³¹ Shafiu Mohammed, PhD,^{232,233} Ali H. Mokdad, PhD,^{6,7} Mohammad Ali Moni, PhD,^{234,235} AmirAli Moodi Ghalibaf, MD,²³⁶ Yousef Moradi, PhD,²³⁷ Paula Moraga, PhD,²³⁸ Rohith Motappa, MD,²³⁹ Admir Mulita, PhD,²⁴⁰ Christopher J. L. Murray, DPhil,^{6,7} Ghulam Mustafa, MD,^{241,242} Ganesh R. Naik, PhD,^{243,244} Zuhair S. Natto, DrPH,^{245,246} Javaid Nauman, PhD,^{247,248} Biswa Prakash Nayak, PhD,²⁴⁹ Nikita Nekliudov, MD,^{6,250} Cuong Tat Nguyen, MPH,²⁵¹ Dang Nguyen, BS,^{252,253} Long Nguyen, PhD,^{254,255} Robina Khan Niazi, PhD,²⁵⁶ Vikram Niranjana, PhD,²⁵⁷ Jean Jacques Noubiap, MD,²⁵⁸ Fred Nugen, PhD,^{259,260} Chimezie Igwegbe Nzopotam, MPH,²⁶¹ Ogochukwu Janet Nzopotam, PhD,^{262,263} Bogdan Oancea, PhD,^{264,265} Ismail A. Odetokun, PhD,²⁶⁶ Osaretin Christabel Okonji, MSc,²⁶⁷ Gláucia Maria Moraes Oliveira, PhD,²⁶⁸ Hany A. Omar, PhD,^{269,270} Michal Ordak, PhD,²⁷¹ Doris Verónica Ortega-Altamirano, EdD,²⁷² Elham H. Othman, PhD,²⁷³ Mahesh P. A, DNB,²⁷⁴ Jagadish Rao Padubidri, MD,²⁷⁵ Ke Pang, MS,^{276,277} Ioannis Pantazopoulos, PhD,^{278,279} Romil R. Parikh, MD,²⁸⁰ Prince Peprah, MSc,²⁸¹ Hoang Nhat Pham, MD,^{197,282} Hoang Tran Pham, MD,²⁸³ Roman V. Polibin, PhD,²⁸⁴ Jagadeesh Puvvula, PhD,²⁸⁵ Jia-Yong Qiu, MD,^{286,287} Mosiur Rahman, DrPH,²⁸⁸ Vahid Rahmanian, PhD,²⁸⁹ Mahmoud Mohammed Ramadan, PhD,^{16,290} Shakthi Kumaran Ramasamy, MD,²⁹¹ Kamleshun Ramphul, MD,²⁹² Ahmed Mustafa Rashid, MD,^{293,294} Salman Rawaf, MD,^{295,296} Elrashdy M. Redwan, PhD,^{297,298} Ana Reis-Mendes, PhD,²⁹⁹ Priyanka Roy, PhD,³⁰⁰ Michele Russo, PhD,³⁰¹ Aly M. A. Saad, MD,³⁰² Maha Mohamed Saber-Ayad, PhD,^{16,303} Cameron John Sabet, MA,¹⁵⁸ Basema Ahmad Saddik, PhD,^{15,304} Mohammad Reza Saeb, PhD,³⁰⁵ Umar Saeed, PhD,^{306,307} Amirhossein Sahebkar, PhD,^{308,309} Mohamed A. Saleh, PhD,^{25,310} Vijaya Paul Samuel, PhD,³¹¹ Abdallah M. Samy, PhD,^{312,313} Elaheh Sanjari, PhD,³¹⁴ Art Schuermans, BSc,^{315,316} Ashenafi Kibret Sendekie, MSc,^{317,318} Subramanian Senthilkumaran, PhD,³¹⁹ Yashendra Sethi, MD,³²⁰ Alfiya Shamsutdinova, MD,³²¹ Vishal Sharma, PhD,³²² Mahabalesh Shetty, MD,³²³ Suraj S Shetty, MD,³²⁴ Ivy Shiue, PhD,^{325,326} Sina Shool, MD,^{152,327} Seyed Afshin Shorofi, PhD,^{328,329} Emmanuel Edwar Siddig, MD,^{330,331} Jasvinder A. Singh, MD,^{332,333} Ranjan Solanki, MD,^{334,335} Prashant Sood, PhD,³³⁶ Soroush Sorane, MD,^{337,338} Michael Spartalis, PhD,³³⁹ Panagiotis Stachteas, MSc,¹⁸³ Joseph R. Starnes, MD,^{340,341} Chandan Kumar Swain, MPhil,³⁴² Lukasz Szarpak, PhD,³⁴³ Sree Sudha T. Y, MD,³⁴⁴ Seyyed Mohammad Tabatabaei, PhD,^{345,346} Celine Tabche, MSc,²⁹⁵ Mohammad Tabish, MPharm,²⁴¹ Mircea Tampa, PhD,^{347,348} Jacques Lukenze Tamuzi, MSc,^{349,350} Mohamad-Hani Temsah, MD,^{351,352} Reem Temsah, PharmD,³⁵³ Masayuki Teramoto, MD,³⁵⁴ Samar Tharwat, MD,³⁵⁵ An Thien Tran, MD,³⁵⁶ Quynh Thuy Huong Tran, PhD,³⁵⁷ Thang Huu Tran, MD,^{358,359} Nguyen Tran Minh Duc, MD,³⁶⁰ Gary Tse, PhD,^{361,362} Lawrence Sena Tuglo, MPH,³⁶³ Munkhtuya Tumurkhuu, PhD,³⁶⁴ Aniefiok John Udoakang, PhD,³⁶⁵ Atta Ullah, MS,³⁶⁶ Era Upadhyay, PhD,³⁶⁷ Jef Van den

Eynde, BSc,³¹⁶ Balachandar Vellingiri, PhD,^{368,369} Maria Viskadourou, MD,³⁷⁰ Loc Tri Vu, MD,¹⁶⁰ Yasir Waheed, PhD,^{371,372} Robert G. Weintraub, MB,^{373,374} Nuwan Darshana Wickramasinghe, MD,³⁷⁵ Suowen Xu, PhD,^{376,377} Amir Yarahmadi, PhD,^{378,379} Naohiro Yonemoto, PhD,^{380,381} Fathiah Zakham, PhD,³⁸² Burhan Abdullah Zaman, PhD,³⁸³ Iman Zare, BSc,³⁸⁴ Michael Zastrozhin, PhD,^{385,386} Zhiqiang Zhang, MD,³⁸⁷ Jiayan Zhou, PhD,³⁸⁸ Magdalena Zielińska, MPharm,³⁸⁹ Jae Il Shin, MD,³⁹⁰ Dong Keon Yon, MD,³⁹¹ Leslie Trumbull Cooper, MD,³⁹² Gregory A Roth, MD^{6,7,393}

Affiliations. ¹Department of Medicine, Kyung Hee University College of Medicine, Seoul, South Korea; ²Center for Digital Health, Medical Science Research Institute, Kyung Hee University, Seoul, South Korea; ³Cardiovascular Disease Initiative, Broad Institute of MIT and Harvard, Cambridge, MA, USA; ⁴Massachusetts General Hospital, Boston, MA, USA; ⁵Department of Pediatric Cardiology, Tehran University of Medical Sciences, Tehran, Iran; ⁶Institute for Health Metrics and Evaluation, University of Washington, Seattle, WA, USA; ⁷Department of Health Metrics Sciences, School of Medicine, University of Washington, Seattle, WA, USA; ⁸Department of Nursing, Al Zaytoonah University of Jordan, Amman, Jordan; ⁹Department of Clinical Governance and Quality Improvement, Aleta Wondo General Hospital, Aleta Wondo, Ethiopia; ¹⁰Centre for Regenerative Medicine and Health, Chinese Academy of Sciences, Hong Kong, China; ¹¹Advanced Diagnostic and Interventional Radiology Research Center (ADIR), Tehran University of Medical Sciences, Tehran, Iran; ¹²Community and Maternity Nursing Unit, University of Duhok, Duhok, Iraq; ¹³Department of Internal Medicine, Federal Medical Centre, Abuja, Nigeria; ¹⁴Department of Family and Community Health, University of Health and Allied Sciences, Ho, Ghana; ¹⁵School of Population Health, University of New South Wales, Sydney, NSW, Australia; ¹⁶Clinical Sciences Department, University of Sharjah, Sharjah, United Arab Emirates; ¹⁷Department of Biopharmaceutics and Clinical Pharmacy, University of Jordan, Amman, Jordan; ¹⁸Department of Pharmacology and Therapeutics, United Arab Emirates University, Al Ain, United Arab Emirates; ¹⁹College of Pharmacy, University of Jordan, Amman, Jordan; ²⁰Department of Biochemistry, Osun State University, Osogbo, Nigeria; ²¹Department of Health Administration and Education, University of Education Winneba, Winneba, Ghana; ²²Department of Epidemiology and Biostatistics, University of Health and Allied Sciences, Ho, Ghana; ²³School of Medicine

and Psychology, Australian National University, Canberra, ACT, Australia; ²⁴Health Research Institute, University of Canberra, Canberra, NSW, Australia; ²⁵College of Medicine, University of Sharjah, Sharjah, United Arab Emirates; ²⁶Institute of Endemic Diseases, University of Khartoum, Khartoum, Sudan; ²⁷Pan-African One Health Institute (PAOHI), Kigali, Rwanda; ²⁸Department of Biosciences, COMSATS Institute of Information Technology, Islamabad, Pakistan; ²⁹Department of Public Health Epidemiology, Debre Berhan University, Debre Birhan, Ethiopia; ³⁰Menelik II Medical and Health Science College, EpiMetrics, Inc., Addis Ababa, Ethiopia; ³¹Brody School of Medicine, East Carolina University, Greenville, NC, USA; ³²Department of Communicable Diseases, Ministry of Health, Muscat, Oman; ³³Middle East, Eurasia, and Africa Influenza Stakeholders Network, Muscat, Oman; ³⁴Ajman University, Ajman, United Arab Emirates; ³⁵Division of Pediatric Cardiology, University of Colorado, Aurora, CO, USA; ³⁶Heart Center, King Faisal Specialist Hospital & Research Center, Riyadh, Saudi Arabia; ³⁷College of Nursing, Qatar University, Doha, Qatar; ³⁸Department of Cardiology, Fudan University, Shanghai, China; ³⁹Department of Physical Therapy and Rehabilitation Sciences, Jordan University of Science and Technology, Irbid, Jordan; ⁴⁰Department of Rehabilitation Sciences and Physical Therapy, Jordan University of Science and Technology, Irbid, Jordan; ⁴¹Department of Emergency Medicine, Sana'a University, Sanaa, Yemen; ⁴²Pediatric Emergency Medicine Department, Drexel University, Philadelphia, PA, USA; ⁴³Department of Family and Community Medicine, University of Jeddah, Jeddah, Saudi Arabia; ⁴⁴Institute of Molecular Biology and Biotechnology (IMBB), The University of Lahore, Lahore, Pakistan; ⁴⁵Faculty of Health Sciences, Equator University of Science and Technology, Uganda, Masaka, Uganda; ⁴⁶Department of Specialty Internal Medicine, Johns Hopkins Aramco Healthcare, Dhahran, Saudi Arabia; ⁴⁷Department of Medicine, Indiana University School of Medicine, Indianapolis, IN, USA; ⁴⁸Department of Pediatric Cardiology, Boston Children's Hospital, Boston, MA, USA; ⁴⁹Department of Pediatrics, Harvard University, Boston, MA, USA; ⁵⁰De Gasperis Cardio Center, Niguarda Hospital, Milan, Italy; ⁵¹School of Medicine and Surgery, University of Milano -Bicocca, Monza, Italy; ⁵²Department of Population and Behavioural Sciences, University of Health and Allied Sciences, Ho, Ghana; ⁵³Department of Medicine, University of Jos, Jos, Nigeria; ⁵⁴Department of Internal Medicine, Jos University Teaching Hospital, Jos, Nigeria; ⁵⁵Department of Cardiology, Carol Davila University

of Medicine and Pharmacy, Bucharest, Romania; ⁵⁶Rural Health Research Institute, Charles Sturt University, Orange, NSW, Australia; ⁵⁷Health Management and Economics Research Center, Iran University of Medical Sciences, Tehran, Iran; ⁵⁸Department of Applied Mathematics, University of Washington, Seattle, WA, USA; ⁵⁹College of Art and Science, Ottawa University, Surprise, AZ, USA; ⁶⁰Department of Biostatistics, Sakarya University, Sakarya, Türkiye; ⁶¹Faculty of Nursing, Philadelphia University, Amman, Jordan; ⁶²Northumbria Health-Care NHS Foundation Trust, Newcastle upon Tyne, UK; ⁶³Institute of Biotechnology and Genetic Engineering, The University of Agriculture, Peshawar, Pakistan; ⁶⁴ASIDE Healthcare, Lewes, DE, USA; ⁶⁵Faculty of Medicine, October 6 University, 6th of October City, Egypt; ⁶⁶Geriatric Unit, Fondazione IRCCS Ca' Granda Ospedale Maggiore Policlinico, Milan, Italy; ⁶⁷Center of Innovation, Technology, and Education (CITE), Anhembí Morumbi University, São José dos Campos, Brazil; ⁶⁸Non-communicable Diseases Research Center, Tehran University of Medical Sciences, Tehran, Iran; ⁶⁹School of Medicine, Iran University of Medical Sciences, Tehran, Iran; ⁷⁰Department of Medical Education, University of Nevada Las Vegas, Las Vegas, NV, USA; ⁷¹Global Health Neurology Lab, NSW Brain Clot Bank, Sydney, NSW, Australia; ⁷²Division of Cerebrovascular Medicine and Neurology, National Cerebral and Cardiovascular Center, Suita, Japan; ⁷³Laboratory of Translational Medicine and Nanotherapeutics, Central University of Punjab, Bathinda, India; ⁷⁴Department of Medical and Surgical Sciences, University of Bologna, Bologna, Italy; ⁷⁵School of Pharmacy, The University of Jordan, Amman, Jordan; ⁷⁶Department of Basic Biomedical Sciences, University of Sharjah, Sharjah, United Arab Emirates; ⁷⁷Center for Nutrition and Health Research, National Institute of Public Health, Cuernavaca, Mexico; ⁷⁸State Disease Investigation Laboratory, Animal Resources Development Department, Agartala, India; ⁷⁹Cardio-Oncology Research Unit, Cardiovascular Analytics Group, Canterbury, UK; ⁸⁰School of Nursing and Health Studies, Hong Kong Metropolitan University, Hong Kong, China; ⁸¹Department of Public Health and Health Sciences, Tennessee State University (TSU), Nashville, TN, USA; ⁸²Department of Community Medicine, Datta Meghe Institute of Medical Sciences, Sawangi, India; ⁸³Department of Cardiology, Peking Union Medical College Hospital, Beijing, China; ⁸⁴Department of Internal Medicine, Peking Union Medical College Hospital, Beijing, China; ⁸⁵Yong Loo Lin School of Medicine, National University of Singapore, Singapore, Singapore; ⁸⁶Division of

Cardiovascular Medicine, Harvard University, Boston, MA, USA; ⁸⁷Department of Scientific Research, Tehran University of Medical Sciences, Tehran, Iran; ⁸⁸Cardiovascular Research Institute, Tehran, Iran; ⁸⁹Division of Infectious Diseases, Virginia Commonwealth University, Richmond, VA, USA; ⁹⁰Department of Medicine, National University of Singapore, Singapore, Singapore; ⁹¹The Interdisciplinary Research Group on Biomedicine and Health, VNU International School (VNUIS), Hanoi, Viet Nam; ⁹²Faculty of Applied Sciences, VNU International School (VNUIS), Hanoi, Viet Nam; ⁹³Life and Health Sciences Research Institute (ICVS), University of Minho, Braga, Portugal; ⁹⁴Institute for Research and Innovation in Health (i3S), University of Porto, Porto, Portugal; ⁹⁵Research Department, Cleveland Clinic Abu Dhabi, Abu Dhabi, United Arab Emirates; ⁹⁶Faculty of Medicine, University of Aleppo, Aleppo, Syria; ⁹⁷Department of Anesthesia, Critical Care and Pain Medicine, Shahid Beheshti University of Medical Sciences, Tehran, Iran; ⁹⁸Center for Immigrant and Refugee Health, Public Health Institute, Los Angeles, CA, USA; ⁹⁹Department of Public Health and Primary Care, University of Cambridge, Cambridge, England; ¹⁰⁰Cardiovascular Metabolic Translational Research Program, National University of Singapore, Singapore, Singapore; ¹⁰¹Department of Anesthesia, Wachemo University, Hossana, Ethiopia; ¹⁰²Chettinad Hospital & Research Institute, Chettinad Academy of Research and Education, Chennai, India; ¹⁰³JSS Medical College, Jagadguru Sri Shivarathreeswara Academy of Health Education and Research, Mysuru, India; ¹⁰⁴Department of Pharmacy, United International University, Dhaka, Bangladesh; ¹⁰⁵Pharmacology Division, Center for Life Sciences Research Bangladesh, Dhaka, Bangladesh; ¹⁰⁶Division of Pathology, ICAR-Indian Veterinary Research Institute, Bareilly, India; ¹⁰⁷Department of Medicine, Can Tho University of Medicine and Pharmacy, Can Tho, Viet Nam; ¹⁰⁸Independent Consultant, Bridgewater, NJ, USA; ¹⁰⁹Department of Cardiology, Hackettstown Medical Center, Hackettstown, NJ, USA; ¹¹⁰Newton Medical Center, Sparta, NJ, USA; ¹¹¹Heart Institute, University of São Paulo, São Paulo, Brazil; ¹¹²Clínica Pró Coração, São Paulo, Brazil; ¹¹³School of Medicine, Federal University of Bahia, Salvador, Brazil; ¹¹⁴Department of Internal Medicine, Escola Bahiana de Medicina e Saúde Pública (Bahiana School of Medicine and Public Health), Salvador, Brazil; ¹¹⁵Department of Infection and Tropical Medicine, University of Sheffield, Sheffield, England; ¹¹⁶School of Nursing and Midwifery, La Trobe University, Melbourne, VIC, Australia; ¹¹⁷Faculty of Science and Health,

University of Portsmouth, Hampshire, UK; ¹¹⁸Division of Cardiovascular Medicine, University of Kentucky, Lexington, KY, USA; ¹¹⁹Division of Cardiology, Harvard University, Boston, MA, USA; ¹²⁰College of Medicine, Korea University, Seoul, South Korea; ¹²¹Houston Methodist Hospital, Houston, TX, USA; ¹²²Basic Medical Sciences Department, University of Sharjah, Sharjah, United Arab Emirates; ¹²³Research Institute of Medical & Health Sciences, University of Sharjah, Sharjah, United Arab Emirates; ¹²⁴Department of Pediatrics, University of Texas, Dallas, TX, USA; ¹²⁵Department of Internal Medicine, Yale University, New Haven, CT, USA; ¹²⁶Department of Epidemiology and Medical Statistics, University of Ibadan, Ibadan, Nigeria; ¹²⁷Research Centre for Healthcare and Community, Coventry University, Coventry, UK; ¹²⁸Tanner College of Dental Medicine, University of Pikeville, Pikeville, KY, USA; ¹²⁹Cardiology Unit, University of Ferrara, Ferrara, Italy; ¹³⁰Department of Social Medicine and Epidemiology, Guilan University of Medical Sciences, Rasht, Iran; ¹³¹Cardiovascular Health and Imaging Laboratory, Centro Nacional de Investigaciones Cardiovasculares (CNIC) (National Centre for Cardiovascular Disease Research), Madrid, Spain; ¹³²Department of Cardiology, Hospital Clinico San Carlos, IDISSC, Madrid, Spain; ¹³³Department of Social Sciences, University of Nicosia, Nicosia, Cyprus; ¹³⁴Institute of Public Health, Charité Universitätsmedizin Berlin (Charité Medical University Berlin), Berlin, Germany; ¹³⁵Department of Community Medicine and Family Medicine, All India Institute of Medical Sciences, Nagpur, India; ¹³⁶School of Medical and Health Science, Edith Cowan University, Australia, Perth, WA, Australia; ¹³⁷Department of Cardiac Surgery, Cleveland Clinic Abu Dhabi, Abu Dhabi, United Arab Emirates; ¹³⁸Lerner College of Medicine, Case Western Reserve University, Cleveland, OH, USA; ¹³⁹Department of Health Systems and Policy Research, Indian Institute of Public Health, Gandhinagar, India; ¹⁴⁰Department of Epidemiology and Biostatistics, Anhui Medical University, Hefei, China; ¹⁴¹Harrington Heart and Vascular Institute, Case Western Reserve University, Cleveland, OH, USA; ¹⁴²Division of Cardiovascular Medicine, Ohio State University, Columbus, OH, USA; ¹⁴³Department of Cardiology, Lehigh Valley Health Network, Allentown, PA, USA; ¹⁴⁴Department of Preventive Cardiology & Medicine, Eternal Heart Care Centre & Research Institute, Jaipur, India; ¹⁴⁵Department of Medicine, Mahatma Gandhi University Medical Sciences, Jaipur, India; ¹⁴⁶Department of Epidemiology and Psychosocial Research, Ramón de la Fuente Muñiz National Institute of Psychiatry, Mexico City,

Mexico; ¹⁴⁷Doctoral Program in Biomedical Gerontology, Pontifical Catholic University of Rio Grande do Sul, Porto Alegre, Brazil; ¹⁴⁸Department of Medical Microbiology, Bahir Dar University, Bahir Dar, Ethiopia; ¹⁴⁹Department of Clinical Pharmacology and Medicine, University of Kufa, Najaf, Iraq; ¹⁵⁰School of Health and Environmental Studies, Hamdan Bin Mohammed Smart University, Dubai, United Arab Emirates; ¹⁵¹Department of Medical and Technical Information Technology, Bauman Moscow State Technical University, Moscow, Russia; ¹⁵²Sina Trauma and Surgery Research Center, Tehran University of Medical Sciences, Tehran, Iran; ¹⁵³Skaane University Hospital, Skaane County Council, Malmö, Sweden; ¹⁵⁴Institute of Pharmaceutical Sciences, University of Veterinary and Animal Sciences, Lahore, Pakistan; ¹⁵⁵Department of Pharmacy Administration and Clinical Pharmacy, Xian Jiaotong University, Xian, China; ¹⁵⁶Department of Neurosurgery, Sichuan University, Chengdu, China; ¹⁵⁷Department of Medicine, MedStar Health, Washington, DC, USA; ¹⁵⁸Department of Medicine, Georgetown University, Washington, DC, USA; ¹⁵⁹Graduate School of Medicine, University of Tokyo, Tokyo, Japan; ¹⁶⁰School of Medicine, Tan Tao University, Long An, Viet Nam; ¹⁶¹Department of Occupational Safety and Health, China Medical University, Taiwan, Taichung, Taiwan; ¹⁶²Department of Cardiovascular Medicine, Mayo Clinic, Phoenix, AZ, USA; ¹⁶³Collaborative Alliance Research and Education (CARE) Programme, Episcopo Research Service, Aberdeen, Scotland; ¹⁶⁴West Africa RCC, Africa Centre for Disease Control and Prevention, Abuja, Nigeria; ¹⁶⁵Department of Community Medicine, University College Hospital, Ibadan, Ibadan, Nigeria; ¹⁶⁶Faculty of Public Health, Universitas Muhammadiyah Aceh (Muhammadiyah University of Aceh), Banda Aceh, Indonesia; ¹⁶⁷School of Pharmacy, BRAC University, Dhaka, Bangladesh; ¹⁶⁸Department of Global Health, South African Medical Research Council, Cape Town, South Africa; ¹⁶⁹Department of Global Health, Stellenbosch University, Cape Town, South Africa; ¹⁷⁰Department of Health and Safety, Dubai Municipality, Dubai, United Arab Emirates; ¹⁷¹Section of Economic & Social Sciences, Humanities & Arts, The World Academy of Sciences UNESCO-TWAS, Trieste, Italy; ¹⁷²Shaanxi University of Technology, Hanzhong, China; ¹⁷³Seoul National University Hospital, Seoul, South Korea; ¹⁷⁴Department of Public Health Sciences, Graduate School of Public Health, Seoul National University, Seoul, South Korea; ¹⁷⁵School of Medicine, Johns Hopkins University, Baltimore, MD, USA; ¹⁷⁶Department of Economics, National Open University, Benin City, Nigeria;

¹⁷⁷Department of Family Medicine and Public Health, University of Opole, Opole, Poland; ¹⁷⁸Department of Oral and Maxillofacial Pathology and Microbiology, Krishna Vishwa Vidyapeeth Deemed to be University, Karad, India; ¹⁷⁹Department of Pharmacology, Post Graduate Institute of Medical Education and Research, Chandigarh, India; ¹⁸⁰Department of Biostatistics, All India Institute of Medical Sciences, Kalyani, India; ¹⁸¹Department of Biostatistics, Indian Council of Medical Research, New Delhi, India; ¹⁸²Harvard T. H. Chan School of Public Health, Harvard University, Boston, MA, USA; ¹⁸³2nd Department of Cardiology, Aristotle University of Thessaloniki, Thessaloniki, Greece; ¹⁸⁴Minimally Invasive Surgery Research Center, Iran University of Medical Sciences, Tehran, Iran; ¹⁸⁵Artificial Intelligence in Health Research Center, Iran University of Medical Sciences, Tehran, Iran; ¹⁸⁶University of California Los Angeles, Los Angeles, CA, USA; ¹⁸⁷Johns Hopkins University, Baltimore, MD, USA; ¹⁸⁸Epidemiology Program, Jazan University, Jazan, Saudi Arabia; ¹⁸⁹BDStatistics Center for Research, Dhaka, Bangladesh; ¹⁹⁰Department of Rehabilitation Sciences, Hong Kong Polytechnic University, Hong Kong, China; ¹⁹¹Family Medicine Department, United Arab Emirates University, Al Ain, United Arab Emirates; ¹⁹²Primary Care Department, NHS North West London, London, UK; ¹⁹³Department of Nursing and Health Promotion, Oslo Metropolitan University, Oslo, Norway; ¹⁹⁴Department of Anthropology, Panjab University, Chandigarh, India; ¹⁹⁵Department of Research Management, Research Institute of Cardiology and Internal Diseases, Almaty, Kazakhstan; ¹⁹⁶Health Research Institute, Al Farabi Kazakh National University, Almaty, Kazakhstan; ¹⁹⁷Department of Cardiovascular Medicine, Mayo Clinic, Rochester, MN, USA; ¹⁹⁸Department of Community Medicine, Rajendra Institute of Medical Sciences, Ranchi, India; ¹⁹⁹Clinical Research Center, Turku University Hospital, Turku, Finland; ²⁰⁰Heart Center, University of Turku and Turku University Hospital, Turku, Finland; ²⁰¹Division of Evidence Synthesis, Foundation for People-centric Health Systems, New Delhi, India; ²⁰²Division of Lifestyle Medicine, Centre for Health: The Specialty Practice, New Delhi, India; ²⁰³Division of Cancer Epidemiology and Genetics, National Cancer Institute, Rockville, MD, USA; ²⁰⁴Department of Precision Medicine, Sungkyunkwan University, Suwon-si, South Korea; ²⁰⁵Department of Cardiothoracic and Vascular Surgery, Westpfalz Klinikum, Kaiserslautern, Germany; ²⁰⁶Department of Cardiothoracic Surgery, University of Patras, Patras, Greece; ²⁰⁷Department of Health Promotion and Health Education, National Taiwan Normal University, Taipei, Taiwan; ²⁰⁸Department of

Medical Sciences, Uppsala University, Uppsala, Sweden; ²⁰⁹Universidad Santa Paula, Curridabat, Costa Rica; ²¹⁰Cardiovascular Pharmacology Unit, Cardiovascular Analytics Group, Hong Kong, China; ²¹¹Department of Clinical Biochemistry and Genetics, Arak University of Medical Sciences, Arak, Iran; ²¹²Nutrition and Food Security Research Center, Isfahan University of Medical Sciences, Isfahan, Iran; ²¹³Internal Medicine Department, Eisenhower Health, Palm Desert, CA, USA; ²¹⁴Research Institute for Endocrine Sciences, Shahid Beheshti University of Medical Sciences, Tehran, Iran; ²¹⁵Australian Centre for Health Services Innovation, Queensland University of Technology, Kelvin Grove, QLD, Australia; ²¹⁶Digital Health and Informatics Directorate, Queensland Health, Brisbane, QLD, Australia; ²¹⁷Clinical Research Development Center, Shiraz University of Medical Sciences, Shiraz, Iran; ²¹⁸Professor Alborzi Clinical Microbiology Research Center, Shiraz University of Medical Sciences, Shiraz, Iran; ²¹⁹Department of Public Health, Arba Minch University, Arba Minch, Ethiopia; ²²⁰Department of Pathology, Imam Abdulrahman Bin Faisal University, Dammam, Saudi Arabia; ²²¹Center for Translation Research and Implementation Science, National Institutes of Health, Bethesda, MD, USA; ²²²Department of Medicine, University of Cape Town, Cape Town, South Africa; ²²³University Centre Varazdin, University North, Varazdin, Croatia; ²²⁴College of Human Medicine, Michigan State University, Flint, MI, USA; ²²⁵Internal Medicine Department, Kyrgyz State Medical Academy, Bishkek, Kyrgyzstan; ²²⁶Department of Atherosclerosis and Coronary Heart Disease, National Center of Cardiology and Internal Disease, Bishkek, Kyrgyzstan; ²²⁷Department of Surgery, Johns Hopkins University, Baltimore, MD, USA; ²²⁸St. Paul's Hospital Millennium Medical College, Addis Ababa, Ethiopia; ²²⁹RAK College of Nursing, RAK Medical and Health Sciences University, Ras Al-Khaimah, United Arab Emirates; ²³⁰Nursing College, Sohag University, Sohag, Egypt; ²³¹Modeling in Health Research Center, Shahrekord University of Medical Sciences, Shahrekord, Iran; ²³²Health Systems and Policy Research Unit, Ahmadu Bello University, Zaria, Nigeria; ²³³Heidelberg Institute of Global Health (HIGH), Heidelberg University, Heidelberg, Germany; ²³⁴AI & Cyber Futures Institute, Charles Sturt University, Bathurst, NSW, Australia; ²³⁵The University of Queensland, Brisbane, QLD, Australia; ²³⁶Mashhad University of Medical Sciences, Mashhad, Iran; ²³⁷Health Metrics and Evaluation Research Center, Kurdistan University of Medical Sciences, Sanandaj, Iran; ²³⁸Computer, Electrical, and Mathematical Sciences and Engineering Division, King

Abdullah University of Science and Technology, Thuwal, Saudi Arabia; ²³⁹Department of Community Medicine, Manipal Academy of Higher Education, Mangalore, India; ²⁴⁰Department of Medicine, Democritus University of Thrace, Alexandroupolis, Greece; ²⁴¹College of Medicine, Shaqra University, Shaqra, Saudi Arabia; ²⁴²Department of Pediatrics & Pediatric Pulmonology, Institute of Mother & Child Care, Multan, Pakistan; ²⁴³College of Medicine and Public Health, Flinders University, Adelaide, SA, Australia; ²⁴⁴Department of Computer Science and IT, Torrens University, Adelaide, SA, Australia; ²⁴⁵Department of Dental Public Health, King Abdulaziz University, Jeddah, Saudi Arabia; ²⁴⁶King Fahd Medical Research Center (KFMRC), Entrepreneurship & Innovation Unit, King Abdulaziz University, Jeddah, Saudi Arabia; ²⁴⁷College of Medicine and Health Sciences, United Arab Emirates University, Al Ain, United Arab Emirates; ²⁴⁸Department of Circulation and Medical Imaging, Norwegian University of Science and Technology, Trondheim, Norway; ²⁴⁹Amity Institute of Forensic Sciences, Amity University, Noida, India; ²⁵⁰Department of Public Health, Erasmus University Medical Center, Rotterdam, Netherlands; ²⁵¹Institute for Global Health Innovations, Duy Tan University, Da Nang, Viet Nam; ²⁵²Harvard T.H. Chan School of Public Health, Harvard University, Cambridge, MA, USA; ²⁵³Department of Medical Engineering, University of South Florida, Tampa, FL, USA; ²⁵⁴Faculty of Public Health, VNU University of Medicine and Pharmacy, Hanoi, Viet Nam; ²⁵⁵International Institute for Training and Research (INSTAR), VNU University of Medicine and Pharmacy, Hanoi, Viet Nam; ²⁵⁶International Islamic University Islamabad, Islamabad, Pakistan; ²⁵⁷School of Medicine, Health Research Institute, University of Limerick, Limerick, Ireland; ²⁵⁸Division of Cardiology, University of California San Francisco, San Francisco, CA, USA; ²⁵⁹Department of Radiology, Mayo Clinic, Rochester, MN, USA; ²⁶⁰School of Information, University of California Berkeley, Berkeley, CA, USA; ²⁶¹Center of Excellence in Reproductive Health Innovation (CERHI), University of Benin, Benin City, Nigeria; ²⁶²Department of Physiology, University of Benin, Edo, Nigeria; ²⁶³Department of Physiology, Benson Idahosa University, Benin City, Nigeria; ²⁶⁴Department of Applied Economics and Quantitative Analysis, University of Bucharest, Bucharest, Romania; ²⁶⁵Bioinformatics Department, National Institute of Research and Development for Biological Sciences, Bucharest, Romania; ²⁶⁶Department of Veterinary Public Health and Preventive Medicine, University of Ilorin, Ilorin, Nigeria; ²⁶⁷School of Pharmacy, University of the Western Cape, Cape Town, South Africa; ²⁶⁸Cardiology Department, Federal University of Rio de Janeiro, Rio de Janeiro, Brazil; ²⁶⁹Department of Pharmacy Practice and Pharmacotherapeutics, University of Sharjah, Sharjah, United Arab Emirates; ²⁷⁰Department of Pharmacology and Toxicology, Beni-Suef University, Beni-Suef, Egypt; ²⁷¹Centre of Regenerative Medicine, Medical University of Bialystok, Bialystok, Poland; ²⁷²Center for Health Systems Research, National Institute of Public Health, Cuernavaca, Mexico; ²⁷³Faculty of Nursing, Applied Science Private University, Amman, Jordan; ²⁷⁴Department of Respiratory Medicine, Jagadguru Sri Shivarathreeswara Academy of Health Education and Research, Mysore, India; ²⁷⁵Department of Forensic Medicine and Toxicology, Manipal Academy of Higher Education, Mangalore, India; ²⁷⁶Department of Neurological Sciences, University of Nebraska Medical Center, Omaha, NE, USA; ²⁷⁷Anesthesiology Department of The Third Xiangya Hospital, Central South University, Changsha, China; ²⁷⁸Department of Emergency Medicine, University of Thessaly, Larissa, Greece; ²⁷⁹Department of Emergency Medicine, University of Bern, Bern, Switzerland; ²⁸⁰Division of Health Policy and Management, University of Minnesota, Minneapolis, MN, USA; ²⁸¹Australian Institute of Health Innovation, Macquarie University, Sydney, NSW, Australia; ²⁸²Department of Internal Medicine, University of Arizona, Tucson, AZ, USA; ²⁸³Department of Internal Medicine, Weiss Memorial Hospital, Chicago, IL, USA; ²⁸⁴Department of Epidemiology and Evidence-Based Medicine, I.M. Sechenov First Moscow State Medical University, Moscow, Russia; ²⁸⁵Department of Biostatistics, Epidemiology, and Informatics, University of Pennsylvania, Philadelphia, PA, USA; ²⁸⁶Guangdong Cardiovascular Institute, Guangdong Provincial People's Hospital, Guangzhou, China; ²⁸⁷Department of Respiratory and Critical Care Medicine, The First Affiliated Hospital, and College of Clinical Medicine of Henan University of Science and Technology, Luoyang, China; ²⁸⁸Department of Population Science and Human Resource Development, University of Rajshahi, Rajshahi, Bangladesh; ²⁸⁹Department of Public Health, Torbat Jam Faculty of Medical Sciences, Torbat Jam, Iran; ²⁹⁰Department of Cardiology, Mansoura University, Mansoura, Egypt; ²⁹¹Department of Radiology, Stanford University, Stanford, CA, USA; ²⁹²Independent Consultant, Las Vegas, NV, USA; ²⁹³Department of Research, Baylor Scott and White Research Institute, Dallas, TX, USA; ²⁹⁴Baylor University, Dallas, TX, USA; ²⁹⁵Department of Primary Care and Public Health, Imperial College London, London, England; ²⁹⁶Academic Public Health

England, Public Health England, London, England; ²⁹⁷Department of Biological Sciences, King Abdulaziz University, Jeddah, Egypt; ²⁹⁸Department of Protein Research, Research and Academic Institution, Alexandria, Egypt; ²⁹⁹Research Centre in Physical Activity, Health and Leisure, University of Porto, Porto, Portugal; ³⁰⁰Department of Labour, Government of West Bengal, Kolkata, India; ³⁰¹Department of Cardiology, SS. Annunziata Hospital - ASL2 Abruzzo, Chieti, Italy; ³⁰²Cardiovascular Department, Zagazig University, Zagazig, Egypt; ³⁰³Department of Medical Pharmacology, Cairo University, Giza, Egypt; ³⁰⁴Department of Public Health and Epidemiology, Khalifa University, Abu Dhabi, United Arab Emirates; ³⁰⁵Department of Pharmaceutical Chemistry, International Medical University, Gdańsk, Poland; ³⁰⁶Széchenyi István University, Széchenyi István University, Győr, Hungary; ³⁰⁷Irfan Suat Gunsul Operational Research Institute, Near East University, Turkey, Nicosia, Türkiye; ³⁰⁸Pharmaceutical Technology Institute, Mashhad University of Medical Sciences, Mashhad, Iran; ³⁰⁹Centre for Research Impact and Outcome, Chitkara College of Pharmacy, Chitkara University, Punjab, India; ³¹⁰Faculty of Pharmacy, Mansoura University, Mansoura, Egypt; ³¹¹Department of Anatomy, Ras Al Khaimah Medical and Health Sciences University, Ras Al Khaimah, United Arab Emirates; ³¹²Department of Entomology, Ain Shams University, Cairo, Egypt; ³¹³Medical Ain Shams Research Institute (MASRI), Ain Shams University, Cairo, Egypt; ³¹⁴Department of Epidemiology and Biostatistics, Tehran University of Medical Sciences, Tehran, Iran; ³¹⁵Faculty of Medicine, Katholieke Universiteit Leuven, Leuven, Belgium; ³¹⁶Department of Cardiovascular Sciences, Katholieke Universiteit Leuven, Leuven, Belgium; ³¹⁷Department of Clinical Pharmacy, University of Gondar, Gondar, Ethiopia; ³¹⁸Curtin School of Diagnostic and Therapeutic Sciences, Curtin University, Bentley, WA, Australia; ³¹⁹Emergency Department, Manian Medical Centre, Erode, India; ³²⁰Department of Medicine, Swami Vivekanand Subharti University, Meerut, India; ³²¹Science Department, Kazakh National Medical University, Almaty, Kazakhstan; ³²²Department of Forensic Science, Panjab University, Chandigarh, India; ³²³Alva's Institute of Medical Sciences & Research Centre, Rajiv Gandhi University of Health Sciences, Moodubidire, India; ³²⁴KS Hegde Medical Academy, Nitte University, Mangalore, India; ³²⁵Oulu Business School, University of Oulu, Oulu, Finland; ³²⁶Martti Ahtisaari Institute, University of Oulu, Oulu, Finland; ³²⁷Center for Technology and Innovation in Cardiovascular Informatics, Iran University of Medical Sciences, Tehran, Iran; ³²⁸Department of Medical-Surgical Nursing, Mazandaran University of Medical Sciences, Sari, Iran; ³²⁹Department of Nursing and Health Sciences, Flinders University, Adelaide, SA, Australia; ³³⁰Unit of Basic Medical Sciences, University of Khartoum, Khartoum, Sudan; ³³¹Department of Medical Microbiology and Infectious Diseases, Erasmus University, Rotterdam, Netherlands; ³³²School of Medicine, Baylor College of Medicine, Houston, TX, USA; ³³³Department of Medicine Service, US Department of Veterans Affairs (VA), Houston, TX, USA; ³³⁴Department of Systemic Pathology, Touro College of Osteopathic Medicine, Middletown, NY, USA; ³³⁵Department of Pathology, American University of the Caribbean School of Medicine, Cupecoy, Saint Martin; ³³⁶Independent Consultant, New Delhi, India; ³³⁷School of Medicine, Urmia University of Medical Sciences, Urmia, Iran; ³³⁸School of Medicine, Babol University of Medical Sciences, Babol, Iran; ³³⁹3rd Department of Cardiology, University of Athens, Athens, Greece; ³⁴⁰Department of Pediatric Cardiology, Vanderbilt University Medical Center, Nashville, TN, USA; ³⁴¹Department of Research and Learning, Lwala Community Alliance, Rongo, Kenya; ³⁴²Department of Analytical and Applied Economics, Utkal University, Bhubaneswar, India; ³⁴³Institute of Medical Sciences, The John Paul II Catholic University of Lublin, Lublin, Poland; ³⁴⁴Department of Pharmacology, All India Institute of Medical Sciences, Deoghar, India; ³⁴⁵Department of Medical Informatics, Mashhad University of Medical Sciences, Mashhad, Iran; ³⁴⁶Applied Biomedical Research Center, Mashhad University of Medical Sciences, Mashhad, Iran; ³⁴⁷Department of Dermatology, Carol Davila University of Medicine and Pharmacy, Bucharest, Romania; ³⁴⁸Department of Dermato-Venereology, Dr. Victor Babes Clinical Hospital of Infectious Diseases and Tropical Diseases, Bucharest, Romania; ³⁴⁹Department of Epidemiology, Stellenbosch University, Cape Town, South Africa; ³⁵⁰Department of Medicine, Northlands Medical Group, Omuthiya, Namibia; ³⁵¹Pediatric Intensive Care Unit, King Saud University, Riyadh, Saudi Arabia; ³⁵²College of Medicine, Alfaisal University, Riyadh, Saudi Arabia; ³⁵³College of Pharmacy, Alfaisal University, Riyadh, Saudi Arabia; ³⁵⁴Department of Preventive Medicine, Northwestern University, Chicago, IL, USA; ³⁵⁵Rheumatology and Immunology Unit, Mansoura University, Mansoura, Egypt; ³⁵⁶Department of Medicine, Pham Ngoc Thach University of Medicine, Ho Chi Minh City, Viet Nam; ³⁵⁷Second Department of Internal Medicine, Kansai Medical University, Hirakata, Japan; ³⁵⁸Department of Internal Medicine, University of Medicine and

Pharmacy at Ho Chi Minh City, Ho Chi Minh City, Viet Nam; ³⁵⁹Department of Business Analytics, University of Massachusetts Dartmouth, Dartmouth, MA, USA; ³⁶⁰Research and Advocacy Initiative, ALS Vietnam, Quang Ngai, Viet Nam; ³⁶¹Department of Population Health, Hong Kong Metropolitan University, Hong Kong, China; ³⁶²Tianjin Key Laboratory of Ionic-Molecular Function of Cardiovascular Disease, Second Hospital of Tianjin Medical University, Tianjin, China; ³⁶³Infectious Disease Epidemiology Research Group, Kwame Nkrumah University of Science and Technology, Kumasi, Ghana; ³⁶⁴Department of Internal Medicine, Wake Forest University, Winston-Salem, NC, USA; ³⁶⁵Department of Biosciences and Biotechnology, University of Medical Sciences, Ondo, Ondo, Nigeria; ³⁶⁶Natural and Medical Sciences Research Center, University of Nizwa, Nizwa, Oman; ³⁶⁷Amity Institute of Biotechnology, Amity University Rajasthan, Jaipur, India; ³⁶⁸Department of Zoology, Central University of Punjab, Bathinda, India; ³⁶⁹Department of Human Genetics & Molecular Biology, Bharathiar University, Coimbatore, India; ³⁷⁰Division of Cardiology, Johns Hopkins University, Baltimore, MD, USA; ³⁷¹Egyetem Square 1, Széchenyi István University, H-9026 Győr, Hungary; ³⁷²Irfan Suat Günsel Operational Research Institute, Near East University, Nicosia, Türkiye; ³⁷³Cardiology Department, Royal Children's Hospital, Melbourne, VIC, Australia; ³⁷⁴Department of Critical Care and Neurosciences, Murdoch Children's Research Institute, Parkville, VIC, Australia; ³⁷⁵Department of Community Medicine, Rajarata University of Sri Lanka, Anuradhapura, Sri Lanka; ³⁷⁶Department of Endocrinology, University of Science and Technology of China, Hefei, China; ³⁷⁷School of Medicine, University of Rochester, Rochester, NY, USA; ³⁷⁸Department of Medicine, Thomas Jefferson University, Philadelphia, PA, USA; ³⁷⁹Department of Medicine, Mashhad University of Medical Sciences, Mashhad, Iran; ³⁸⁰Department of Biostatistics, University of Toyama, Toyama, Japan; ³⁸¹Department of Public Health, Juntendo University, Tokyo, Japan; ³⁸²Faculty of Medicine and Health Sciences, Hodeidah University, Hodeidah, Yemen; ³⁸³Basic Sciences Department, University of Duhok, Duhok, Iraq; ³⁸⁴Research and Development Department, Sina Medical Biochemistry Technologies, Shiraz, Iran; ³⁸⁵Department of Bioengineering and Therapeutical Sciences, University of California San Francisco, San Francisco, CA, USA; ³⁸⁶Department of Administration, PGxAI, San Francisco, CA, USA; ³⁸⁷Tianjin Medical University General Hospital, Tianjin Centers for Disease Control and Prevention, Tianjin, China; ³⁸⁸School of Medicine, Stanford University, Stanford,

CA, USA; ³⁸⁹Department of Biochemistry and Pharmacogenomics, Medical University of Warsaw, Warsaw, Poland; ³⁹⁰Department of Pediatrics, Yonsei University College of Medicine, Seoul, South Korea; ³⁹¹Department of Pediatrics, Kyung Hee University College of Medicine, Seoul, South Korea; ³⁹²Department of Cardiovascular Medicine, Mayo Clinic, Jacksonville, FL, USA; ³⁹³Division of Cardiology, University of Washington, Seattle, WA, USA.

AUTHOR CONTRIBUTIONS Please see the [Supplemental Appendix](#) for more detailed information about individual author contributions to the research, divided into the following categories: providing data or critical feedback on data sources; developing methods or computational machinery; providing critical feedback on methods or results; drafting the paper or revising it critically for important intellectual content; and managing the estimation or publications process. Members of the core research team for this topic area had full access to the underlying data used to generate estimates presented in this study (D.K.Y. and G.A.R.). All other authors had access to and reviewed estimates as part of the research evaluation process, which includes additional stages of formal review. Sooji Lee, Hayeon Lee, and Min Seo Kim contributed equally. Jae Il Shin and Dong Keon Yon contributed equally. Leslie T. Cooper, Jr, and Gregory A. Roth served as senior authors. The corresponding author and senior authors had full access to the data in the study and had the final responsibility for the decision to submit for publication.

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committee or advocacy group, paid or unpaid as chair, World Heart Federation Science Committee and as member, GBD Scientific Committee; stock or stock options in NVDA (Nvidia Corporation), CRWD (CrowdStrike Holdings, Inc), IYU (Ishares U.S. Technologies), and PLTR (Palantir Technologies) outside of the submitted work. Dr Mohammed has received support for the present manuscript from the Gates Foundation outside of the submitted work. Dr Oancea has received support for the present manuscript from UEFISCDI (project PN-IV-P6-6.1-CoEx-2024-0196, contract number 17/2.03.2026) and from MRID (project PNRR-I8 number 842027778, contract number 760096) outside of the submitted work. Dr Sharma has received other financial or nonfinancial interests: support from DFSS (MHA)’s research project (DFSS28(1)2019/EMR/6) at Department of Forensic Science (Formerly Institute of Forensic Science & Criminology), Panjab University, Chandigarh, India, RUSA 2.0 grant to Panjab University and PAIR grant (ANRF/PAIR/2025/000006/PAIR) to Panjab University from ANRF outside of the submitted work. Dr Shin has received support for this work from the Yonsei Fellowship, funded by Lee Youn Jae outside of the submitted work. Dr Singh has received consulting fees from ROMTech, Atheneum, Clearview health care partners, Yale, Hulo, Horizon Pharmaceuticals/DINORA, ANI/Exeltis USA Inc, SOBI Pharmaceuticals, Frictionless Solutions, Schipher, Crealta/Horizon, Medisys, Fidia, PK Med, Two labs Inc, Adept Field Solutions, Clinical Care options, Putnam Associates, Focus Forward, Navigant Consulting, Spherix, MedIQ, Jupiter Life Science, UBM LLC, Trio Health, Medscape, WebMD, and Practice Point communications; the National Institutes of Health; and the American College of Rheumatology (consultant fees paid to Dr Singh for each entity); has received payment or honoraria for lectures, presentations, speakers bureaus, manuscript writing or educational events: Dr Singh is on the Speakers Bureau of Simply Speaking (he previously served as a consultant and consultant fees were paid to him); has received support for attending meetings and/or travel from past steering committee member of OMERACT (he previously received support from the organization to attend their meeting every 2 years); participates on a Data Safety Monitoring Board or Advisory Board with the FDA Arthritis Advisory Committee (he served previously as a member; no financial support); has a leadership or fiduciary role in other board, society, committee or advocacy group, paid or unpaid as past steering committee member of the OMERACT, an international organization that develops measures for clinical trials and receives arm’s length funding from 12 pharmaceutical companies (he previously received support from the organization to attend their meeting every 2 years); has stock or stock options in Atyr pharmaceuticals, Atai Life Sciences, Kintara therapeutics, Intelligent Biosolutions, Acumen pharmaceutical, TPT Global Tech, Vaxart pharmaceuticals, Atyu Biopharma, Adaptimmune Therapeutics, GeoVax Labs, Pieris Pharmaceuticals, EnzoLytics Inc, Seres Therapeutics, Tonix Pharmaceuticals Holding Corporation, Aebona Pharmaceuticals, and Charlotte’s Web Holdings, Inc (he owns stock options), and in Amarin, Viking, and Moderna pharmaceuticals (he previously owned stock options) outside of the submitted work. Dr Starnes has participated on a Data Safety Monitoring Board or Advisory Board with Medtronic (has received payment for serving on advisory board for pulse oximetry) outside of the submitted work. Dr Upadhyay has patents planned, issued or pending: a system and method of reusable filters for antipollution mask, a system and method for electricity generation through crop stubble by using microbial fuel cells, a system for disposed personal protection equipment (PPE) into biofuel through pyrolysis and method, a novel herbal pharmaceutical aid for formulation of gel and method thereof, herbal drug formulation for treating lung tissue degenerated by particulate matter exposure, and a method to transform cow dung into the wall paint by using natural materials and composition thereof (all published); and biodegradable packaging composition and method of preparation thereof, eco-friendly bio-shoe polish from banana and turmeric, honey-based polyherbal syrup composition to treat air pollution-induced inflammation and preparation

method thereof, process for preparing a caffeine free, antioxidant and nutrient rich beverage, polyherbal oral mist composition for dual relief and method for preparation thereof, and method for fabricating sustainable waterproof material sheets from LDPE plastic wastes (all filed); leadership or fiduciary role in other board, society, committee or advocacy group, paid or unpaid as Former Joint Secretary, Executive Council Member, Indian Meteorological Society, Jaipur Chapter (India); and as member secretary of the DSTPURSE program outside of the submitted work. Dr Weintraub has participated on a Data Safety Monitoring Board or Advisory Board with the VALOR study advisory board (pediatric heart failure drug trial) outside of the submitted work. Dr Zielińska has other financial or

nonfinancial interests as an Alexion AstraZeneca Rare Disease employee outside of the submitted work. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr Dong Keon Yon, Department of Pediatrics, Kyung Hee University Medical Center, Kyung Hee University, 23 Kyunghee-dae-ro, Dongdaemun-gu, Seoul 02447, South Korea. E-mail: yonkkang@gmail.com.

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KEY WORDS DALY, Global Burden of Disease, Injuries, and Risk Factors Study, impairment, incidence, mortality, myocarditis, prevalence

APPENDIX For an expanded Methods section as well as supplemental tables, figures, and references, please see the online version of this paper.